

Satellite communications

-Orbital Mechanics

ORBITAL MECHANICS

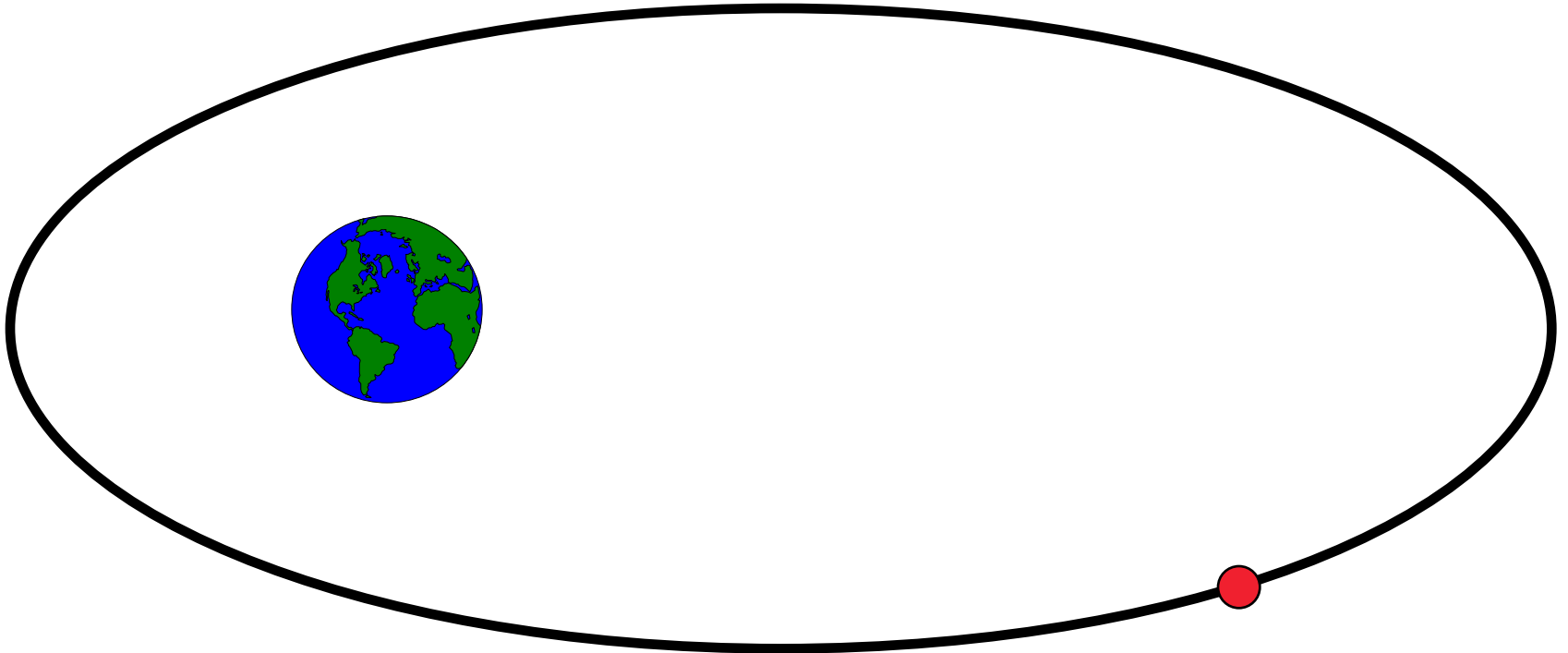
- Physical Laws
- Requirements for Injection
- Classifications of Orbit
- Six Orbital Elements
- Ground Tracks

ORBITAL MECHANICS

- Two men in history that were essential to formulating orbital mechanics:
Kepler and Newton!!
- Kepler's 3 Laws:
 - Law of Ellipses
 - Law of Equal areas
 - Law of Harmonics
- Newton's 3 Laws:
 - Law of Inertia
 - Law of Momentum
 - Law of Action -Reaction

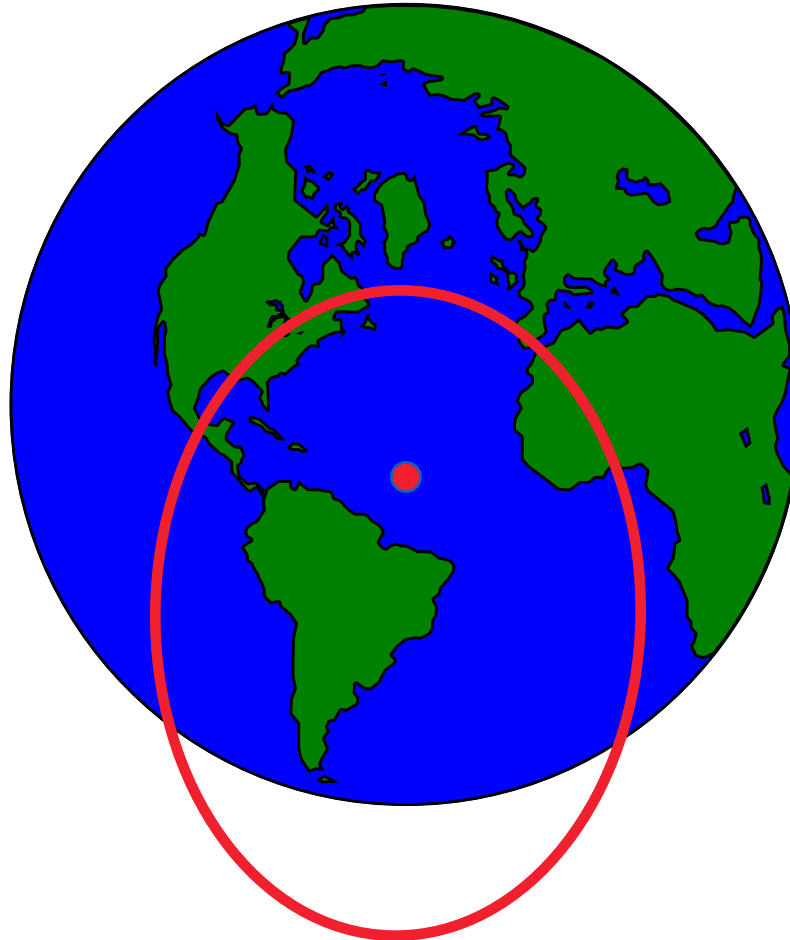
PHYSICAL LAWS

Kepler's 1st Law: Law of Ellipses



The orbits of the planets are ellipses with the sun at one focus.
Or, the orbits of satellites around the earth are ellipses with
the earth at one focus.....

PHYSICAL LAWS

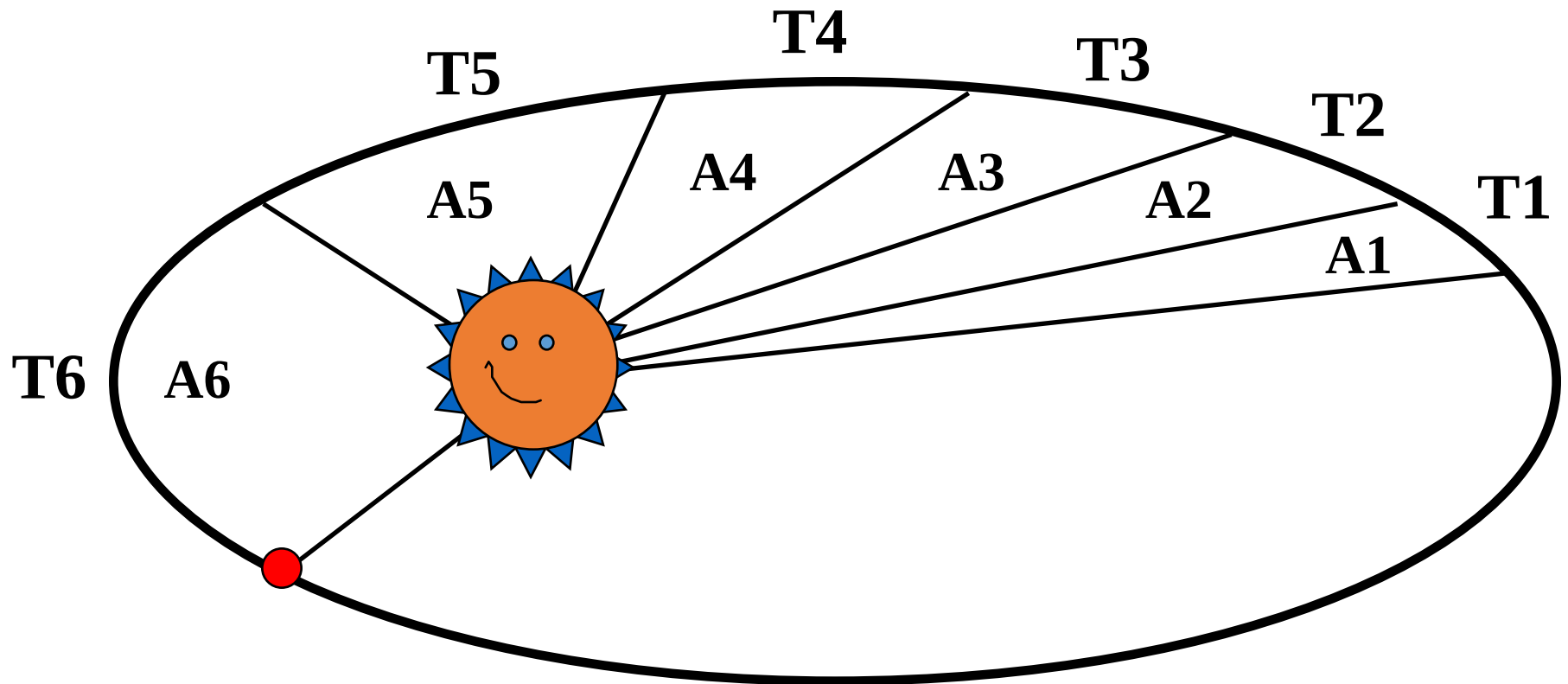


Is this orbit possible?

PHYSICAL LAWS

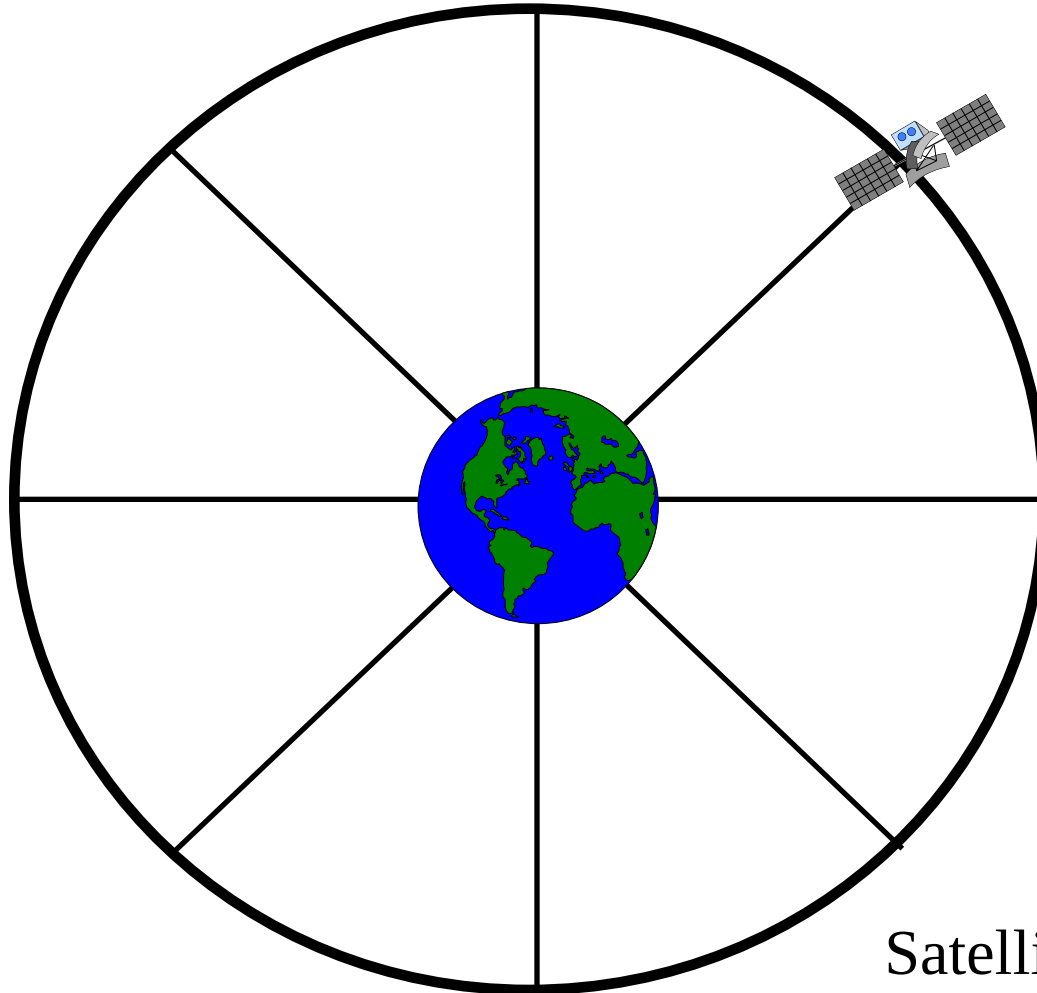
Kepler's 2nd Law: Law of Equal Areas

The line joining the planet to the center of the sun sweeps out equal areas in equal times



PHYSICAL LAWS

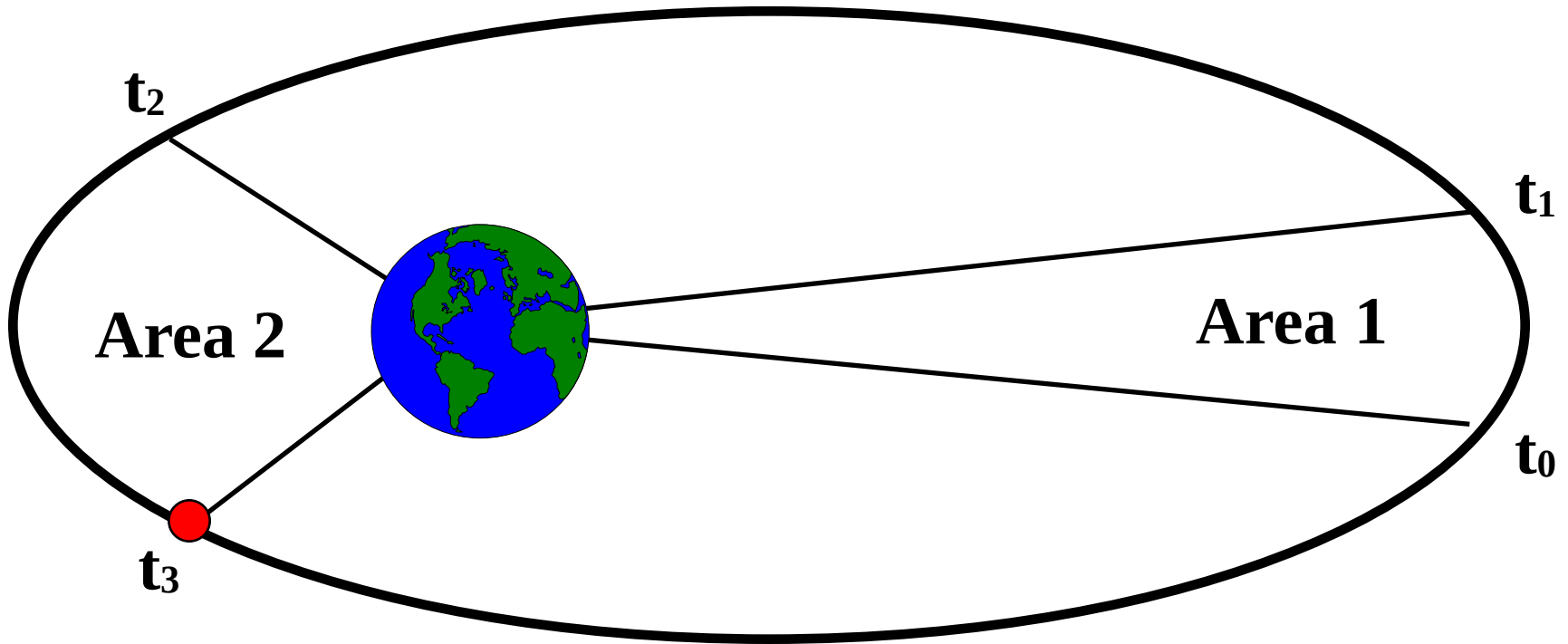
Kepler's 2nd Law: Law of Equal Areas



Satellites travel at the
same speed!!

PHYSICAL LAWS

Kepler's 2nd Law: Law of Equal Areas



$$t_1 - t_0 = t_3 - t_2$$

$$\text{Area 1} = \text{Area 2}$$

Satellites travel at
varying speeds!!

PHYSICAL LAWS

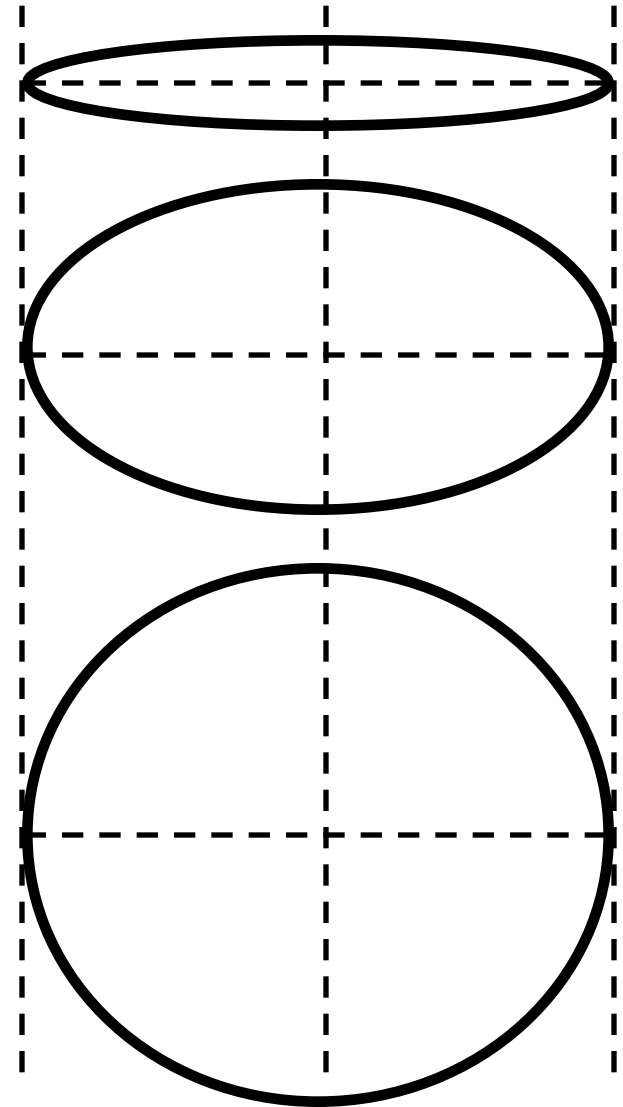
Kepler's 3rd Law: Law of Harmonics

The squares of the periods of
two planets' orbits are
proportional to each other as
the cubes of their semi-major
axes:

$$T_1^2/T_2^2 = a_1^3/a_2^3$$

In English:

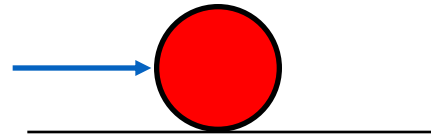
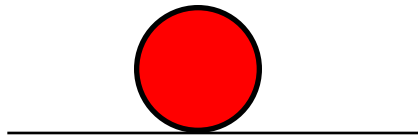
Orbits with the same semi-
major axis will have the same
period



PHYSICAL LAWS

Newton's 1st Law: Law of Inertia

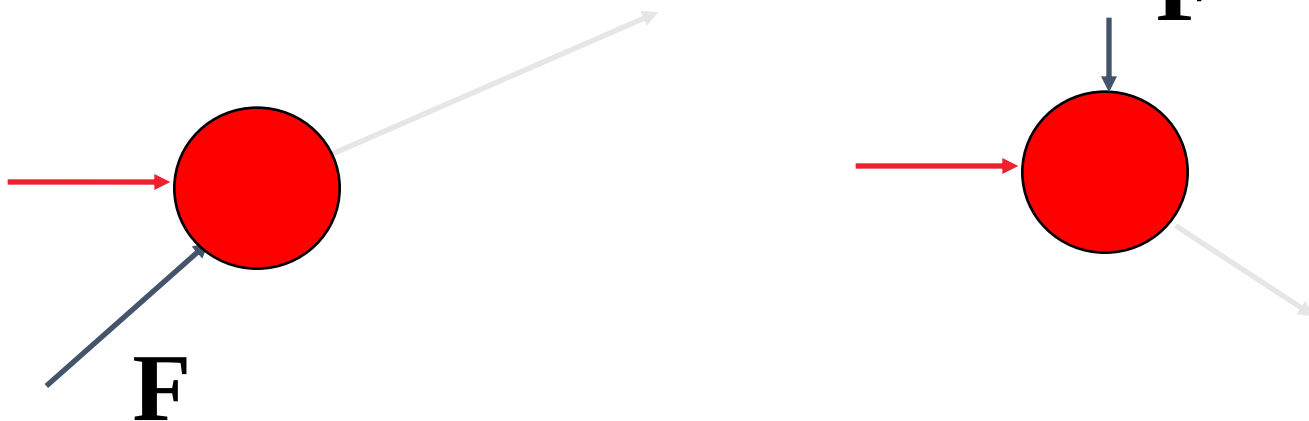
- Every body continues in a state of uniform motion unless it is compelled to change that state by a force imposed upon it



PHYSICAL LAWS

Newton's 2nd Law: Law of Momentum

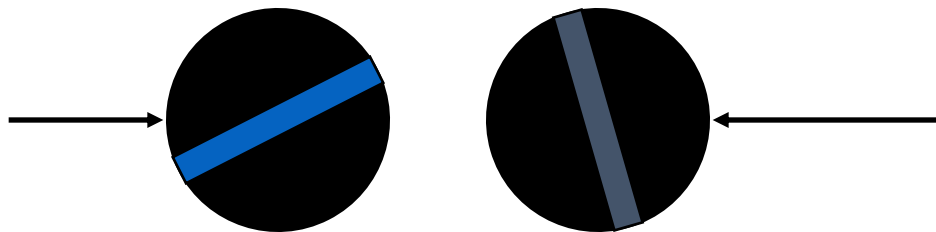
- Change in momentum is proportional to and in the direction of the force applied
- Momentum equals mass x velocity
- Change in momentum gives: $F = ma$



PHYSICAL LAWS

Newton's 3rd Law: Action - Reaction

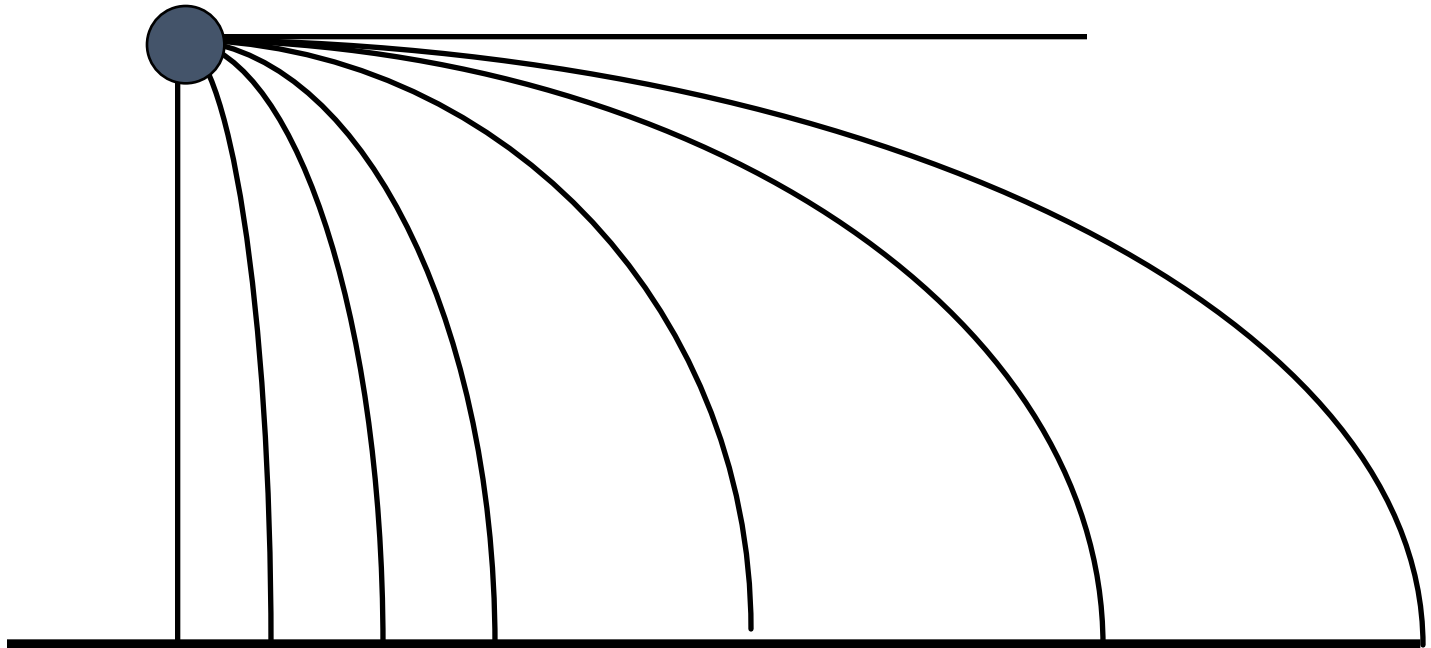
- For every action, there is an equal and opposite reaction
- Hints at conservation of momentum



INJECTION REQUIREMENTS

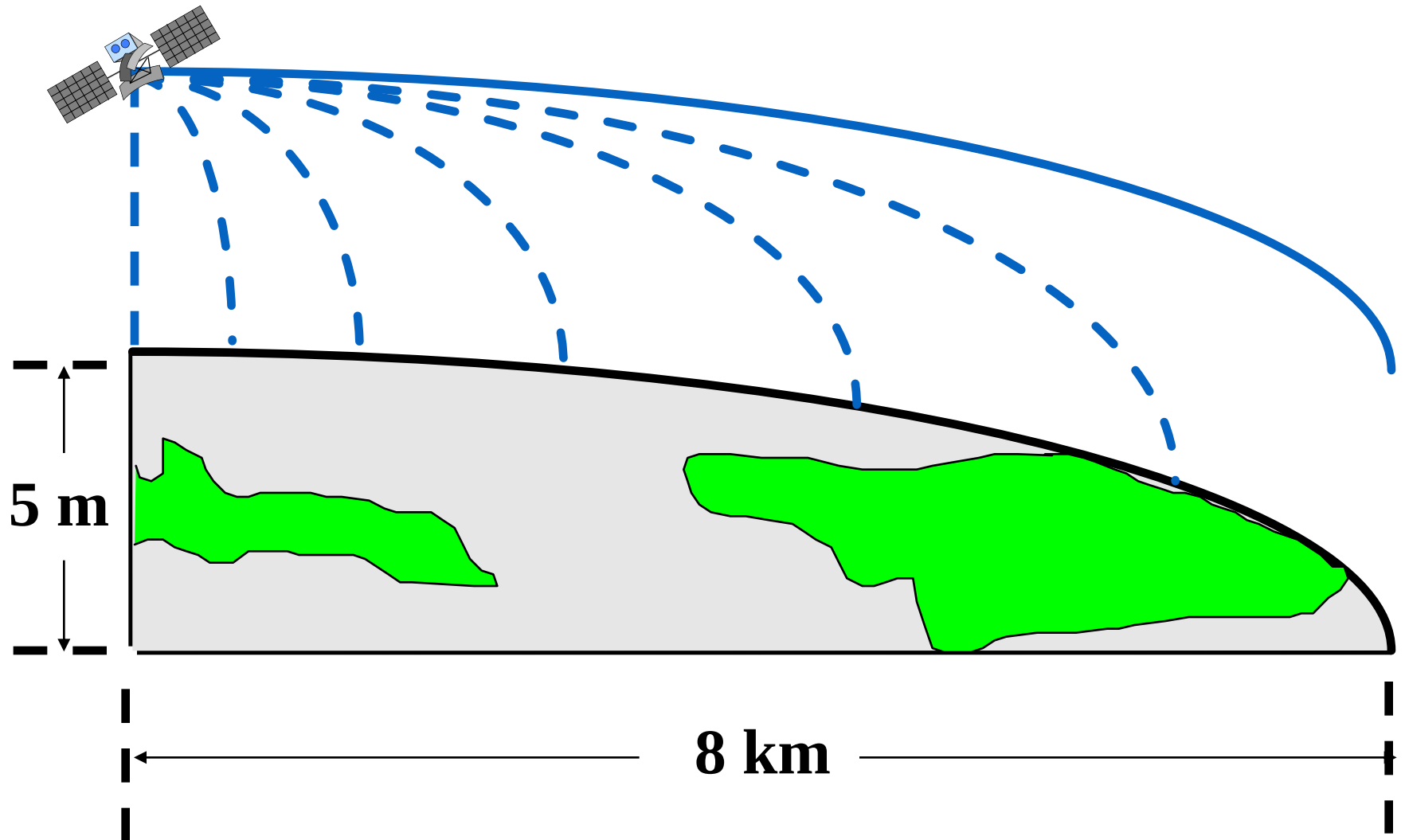
Speed

If you want something to stay in an orbit, it has to be going very fast!



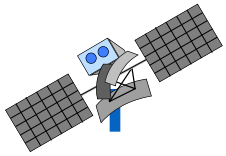
INJECTION REQUIREMENTS

Speed

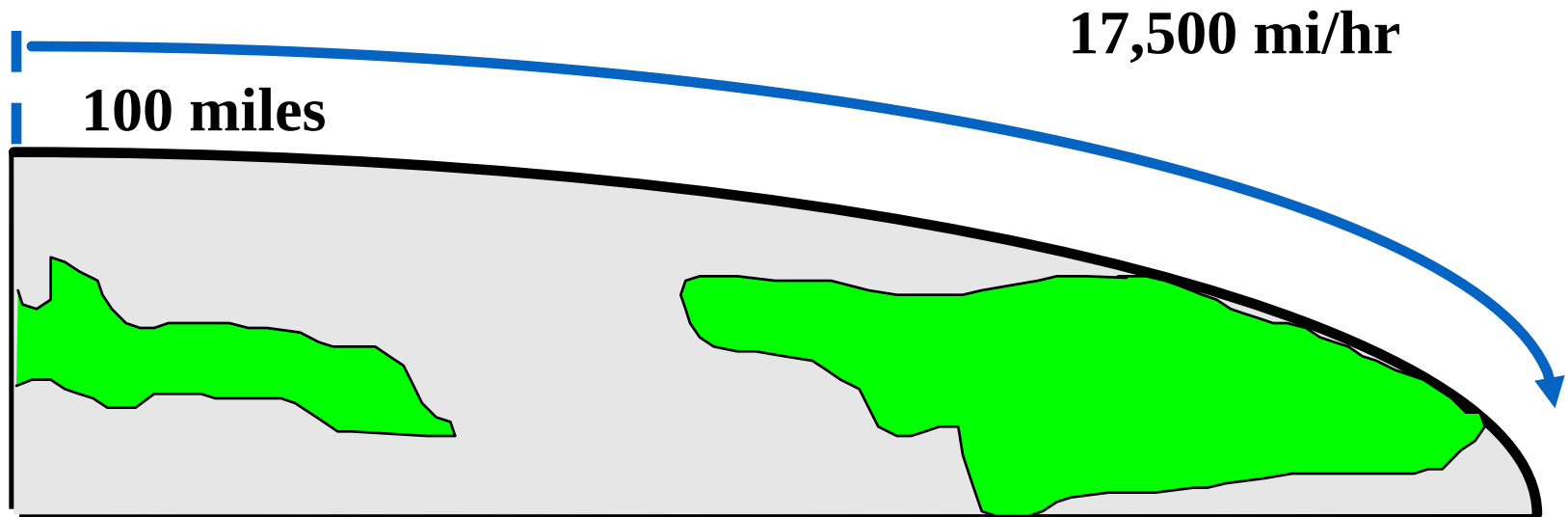


INJECTION REQUIREMENTS

Speed

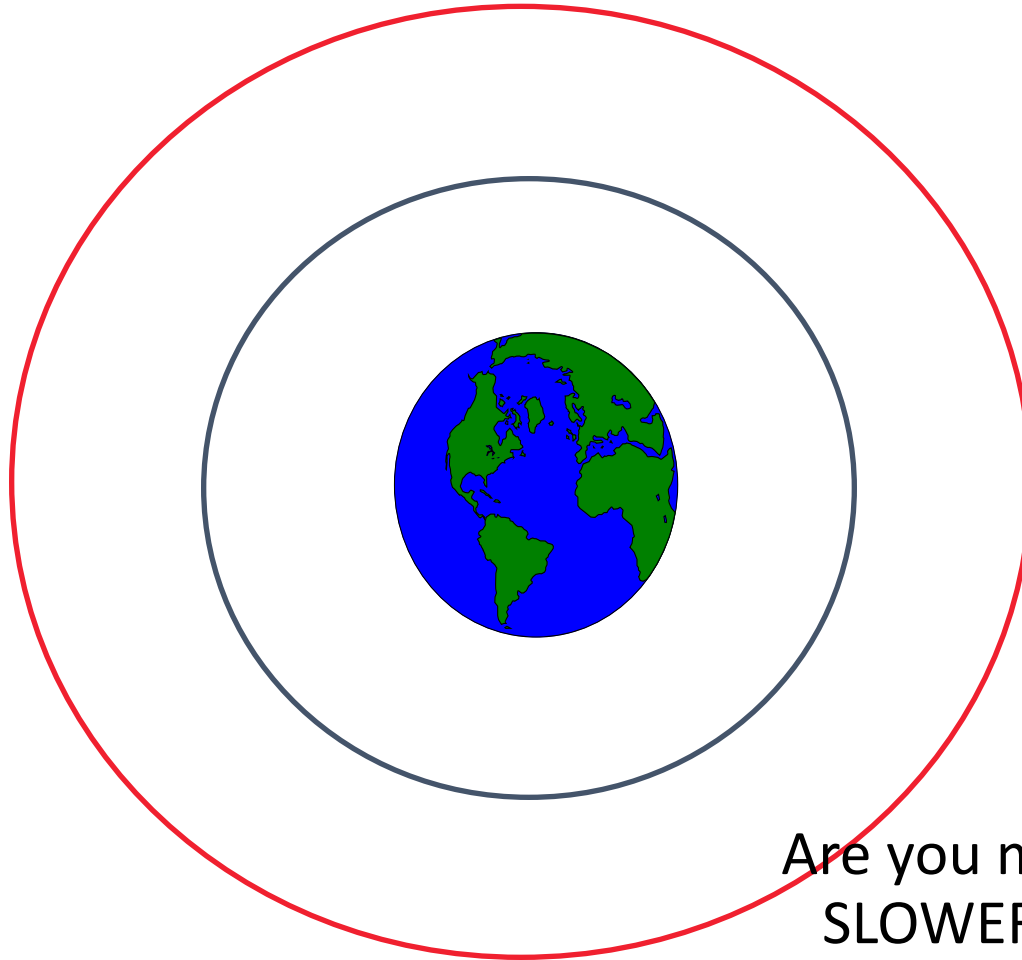


A satellite must be going 17,500 mph to stay in a low earth orbit



INJECTION REQUIREMENTS

Altitude

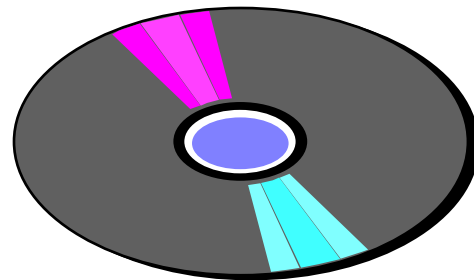
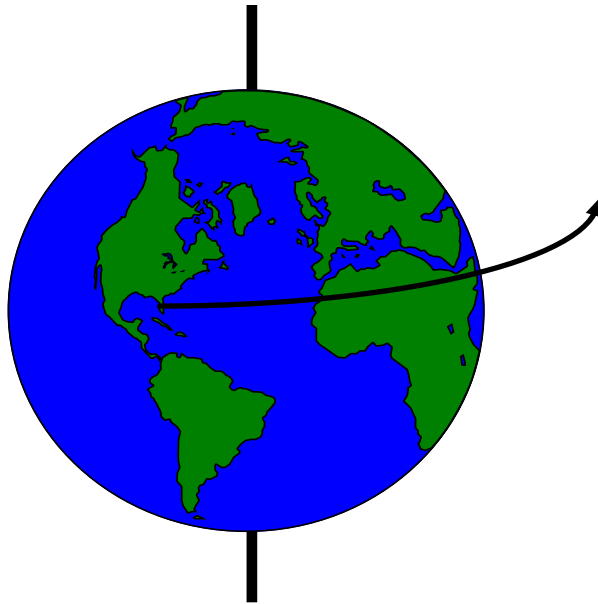


Are you moving FASTER or
SLOWER the higher your
altitude?

INJECTION REQUIREMENTS

Direction

- Since the earth rotates from west to east, you want to launch satellites to the east
- This give you a 915 mph speed boost by launching east (at the Kennedy Space Center's location in Florida)



What happens if you launch to the west? The south?

ORBITAL ELEMENTS

Definition

- A set of mathematical parameters that enables us to accurately describe satellite motion

ORBITAL ELEMENTS

Purpose

- Discriminate one satellite from other satellites
- Predict where a satellite will be in the future or has been in the past
- Determine amount and direction of maneuver or perturbation

ORBITAL ELEMENTS

or

The Six Keplerian Elements

- Size/Period
- Shape (Circular or Ellipse)
eccentricity
- Inclination
- Right Ascension
- Argument of Perigee
- True Anomaly

ORBIT CLASSIFICATION

Size/Period

- Size is how big or small your satellite's orbit is....
- Defined by semi-major axis
- There are basically 4 sizes of orbits satellites use:
 - Low Earth Orbit (LEO): approx 120 – 1200 miles above Earth
 - Medium Earth Orbit (MEO) or Semi-synchronous Orbit: approx 12,000 miles above Earth
 - Highly Elliptical Orbit (HEO): altitude varies greatly! From 100 miles to sometimes several hundred thousand miles
 - Geo-synchronous or Geo-stationary Orbit (GEO): approx 22,300 miles from Earth

ORBIT CLASSIFICATION

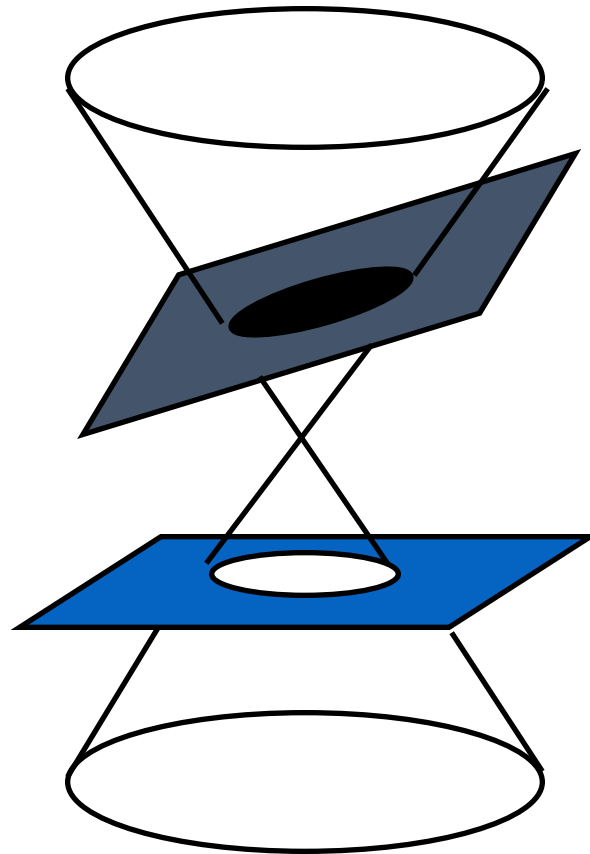
Location of Orbits

- Equatorial – Prograde (towards the east) or Retrograde (towards the west)
- Polar – Over the Poles!!
- A very Important Point:
ALL ORBITS OF SATELLITES MUST INTERSECT THE
CENTER OF THE EARTH

ORBIT CLASSIFICATION

Shape

Orbit shapes are either circular or not circular: some sort of an Ellipse!!
How elliptical an orbit, is called Eccentricity



ORBIT CLASSIFICATIONS

Circular Orbits

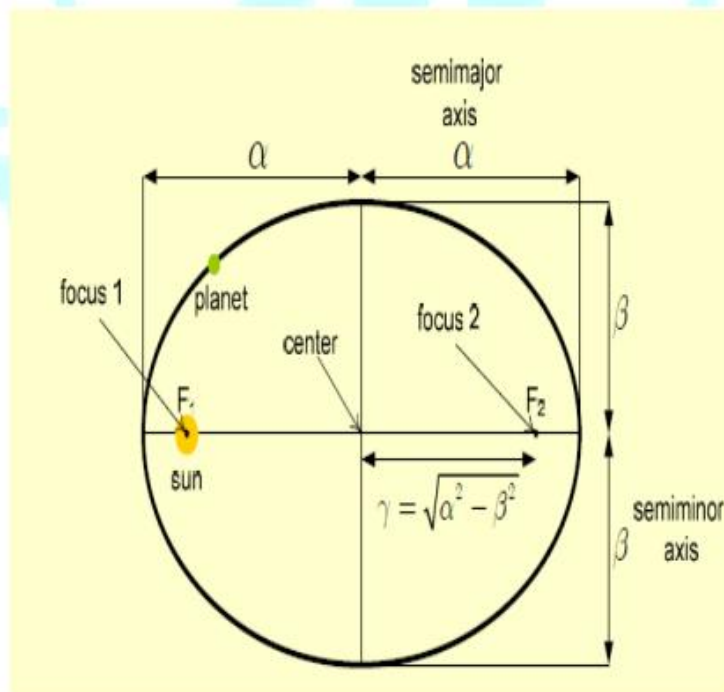
- Characteristics
 - Constant speed
 - Nearly constant altitude
- Typical Missions
 - Reconnaissance/Weather (DMSP)
 - Manned
 - Navigational (GPS)
 - Geo-synchronous (Comm sats)

ORBIT CLASSIFICATIONS

Elliptical Orbits

- Characteristics
 - Varying speed
 - Varying altitude
 - Asymmetric Ground Track
- Typical Missions
 - Deep space surveillance (Pioneer)
 - Communications (Polar comm.)
 - Ballistic Missiles

Eccentricity

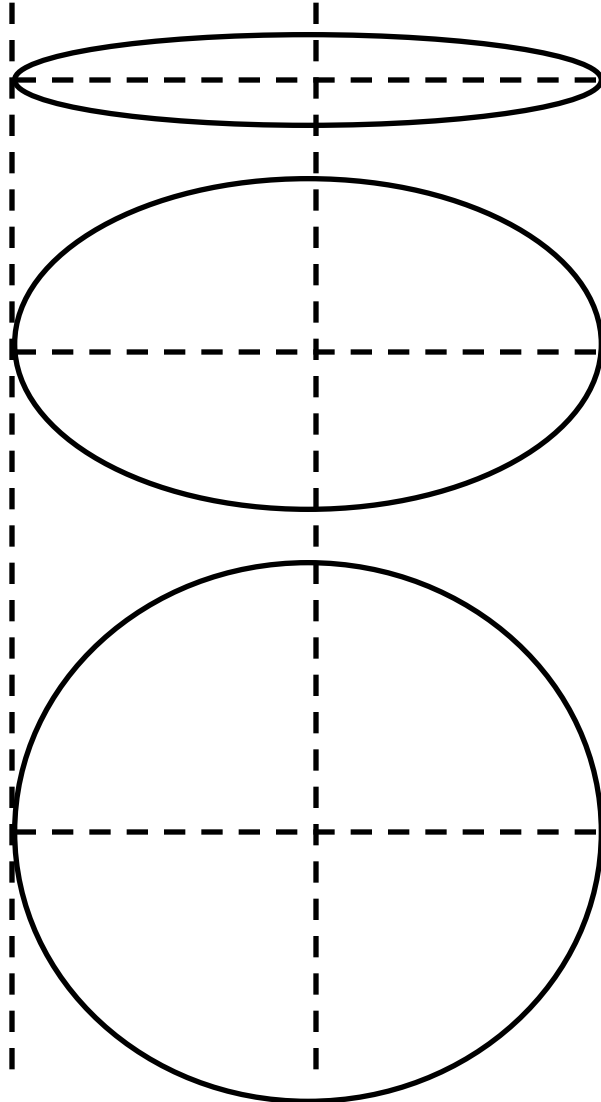


$$\text{eccentricity} = \varepsilon = \frac{\sqrt{a^2 - b^2}}{a} = \frac{\gamma}{a}$$

For an elliptical orbit, $0 < e < 1$. When $e = 0$, the orbit becomes circular.

ORBIT CLASSIFICATIONS

Eccentricity



**The closer
your
Eccentricity is
to 1, the more
elliptical your
orbit is**

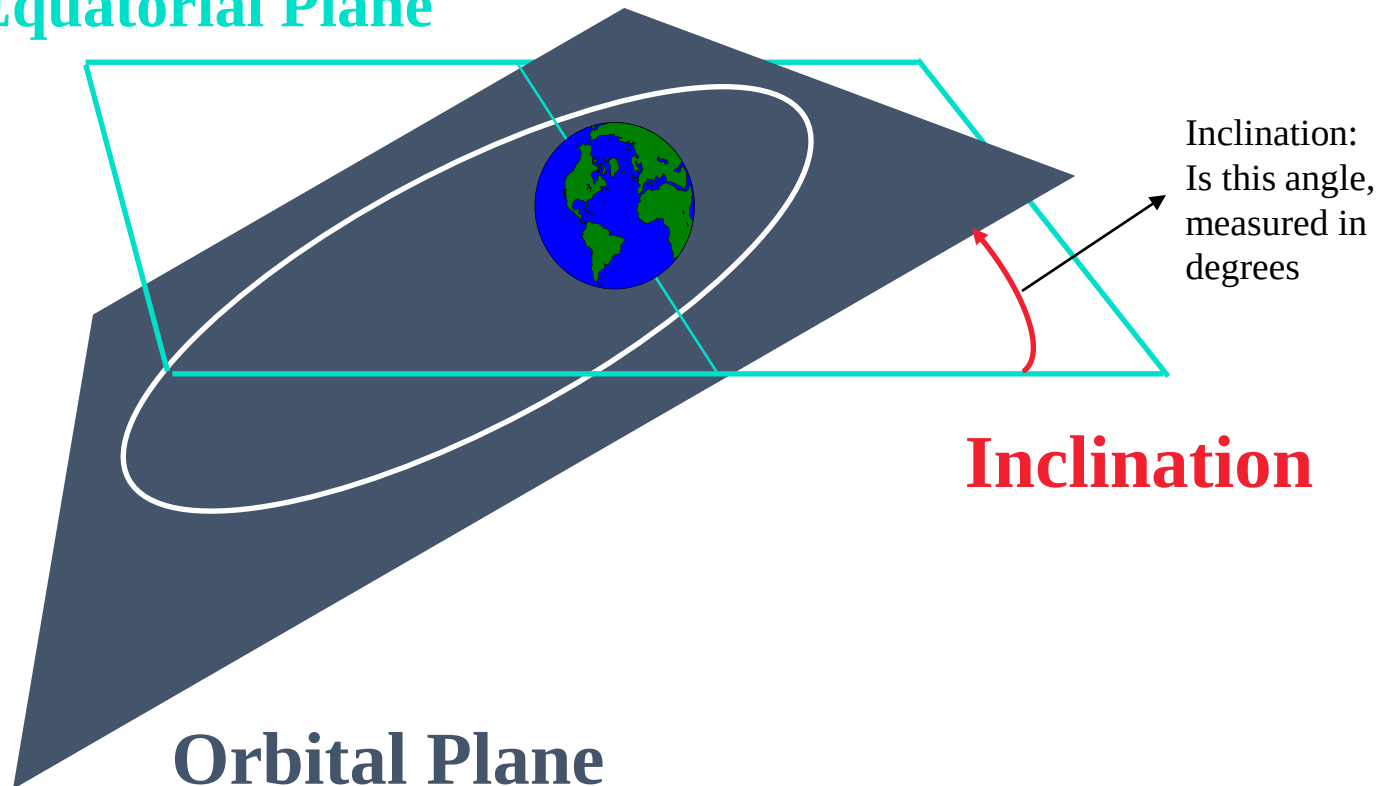
**Why could you never have an
Eccentricity of 1??**

ORBITAL ELEMENTS

Inclination

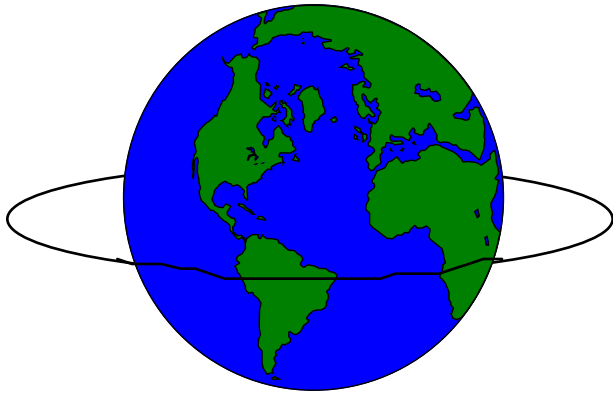
- Inclination is the *tilt* of your orbit
 - At 0 degrees of inclination, you are orbiting the equator
 - At 90 degrees of inclination, you are in a polar orbit

Equatorial Plane

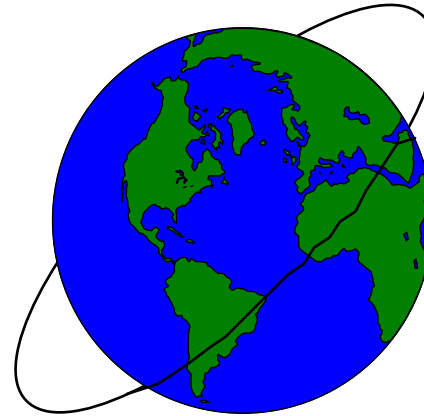


ORBITAL ELEMENTS

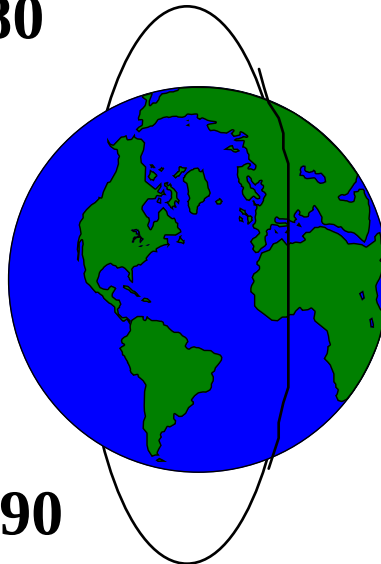
Inclination



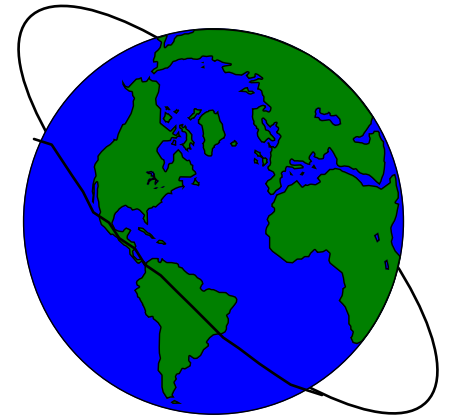
Equatorial: $i = 0$ or 180



**Prograde:
 $0 \leq i < 90$**



Polar: $i = 90$

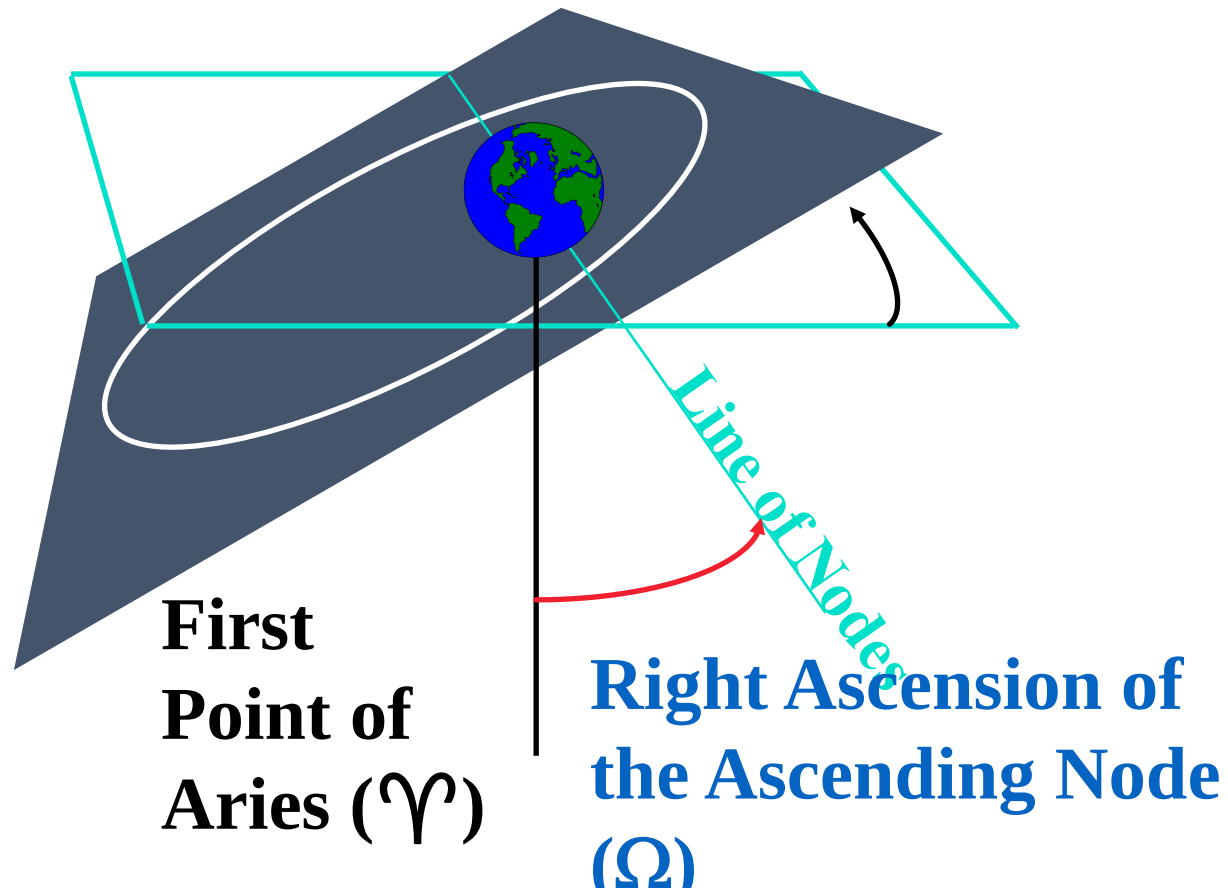


**Retrograde:
 $90 < i \leq 180$**

ORBITAL ELEMENTS

Right Ascension

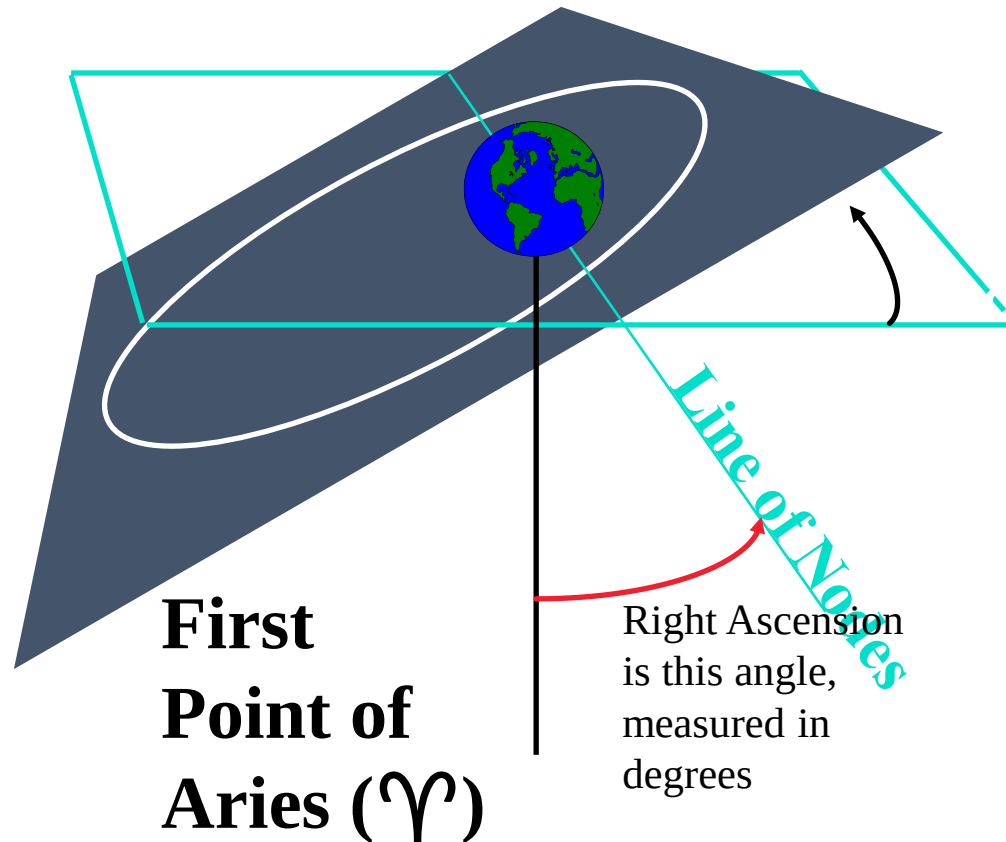
- Right Ascension is the *swivel* of your tilt, as measured from a fixed point in space, called the First Point of Aries



ORBITAL ELEMENTS

Right Ascension

- Right Ascension will determine where your satellite will cross the Equator on the ascending pass
- It is measured in degrees



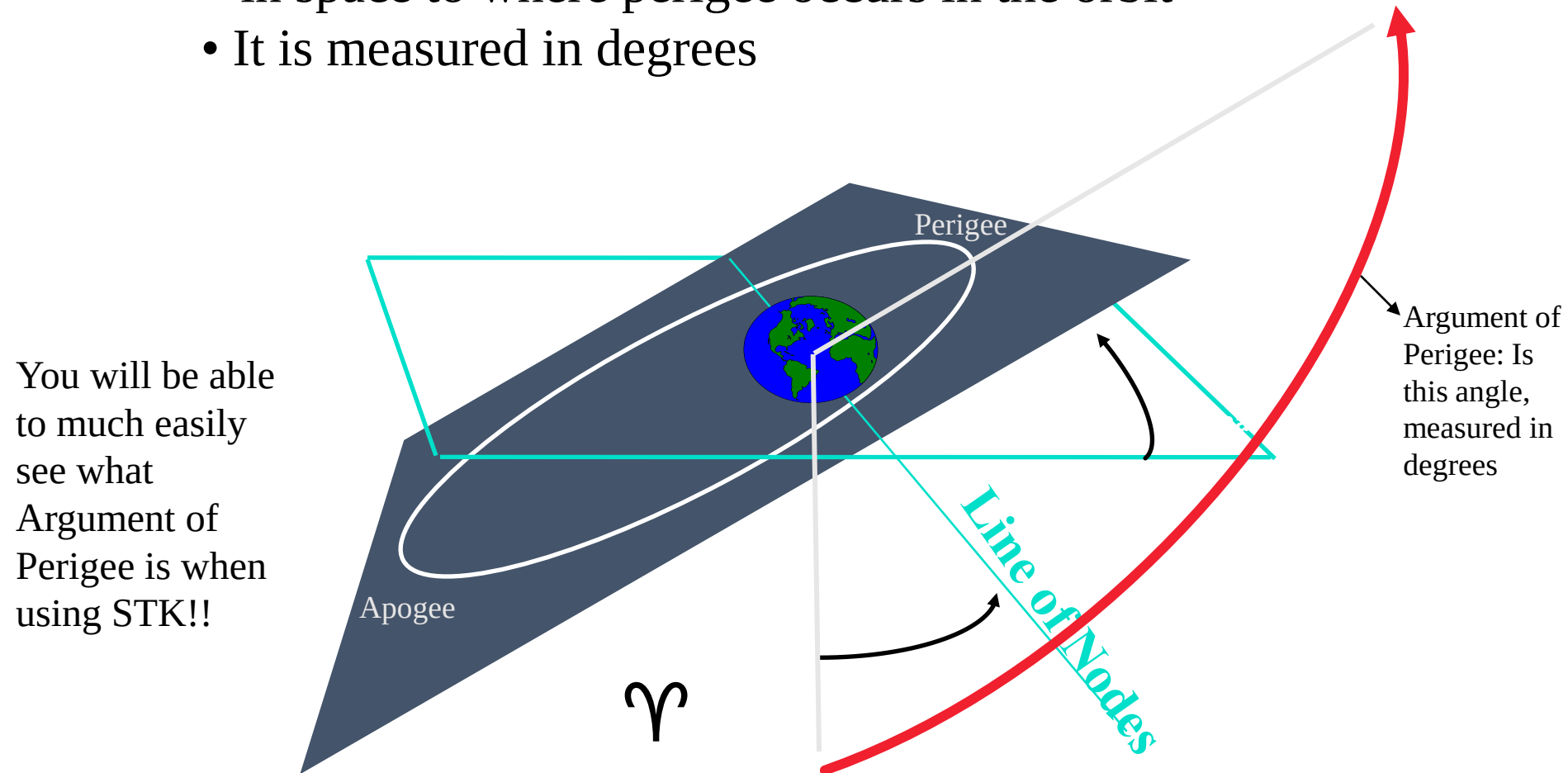
You will be able to much easily see what Right Ascension is when using STK!!

You will not have a Right Ascension if your Inclination is 0, why?

ORBITAL ELEMENTS

Argument of Perigee

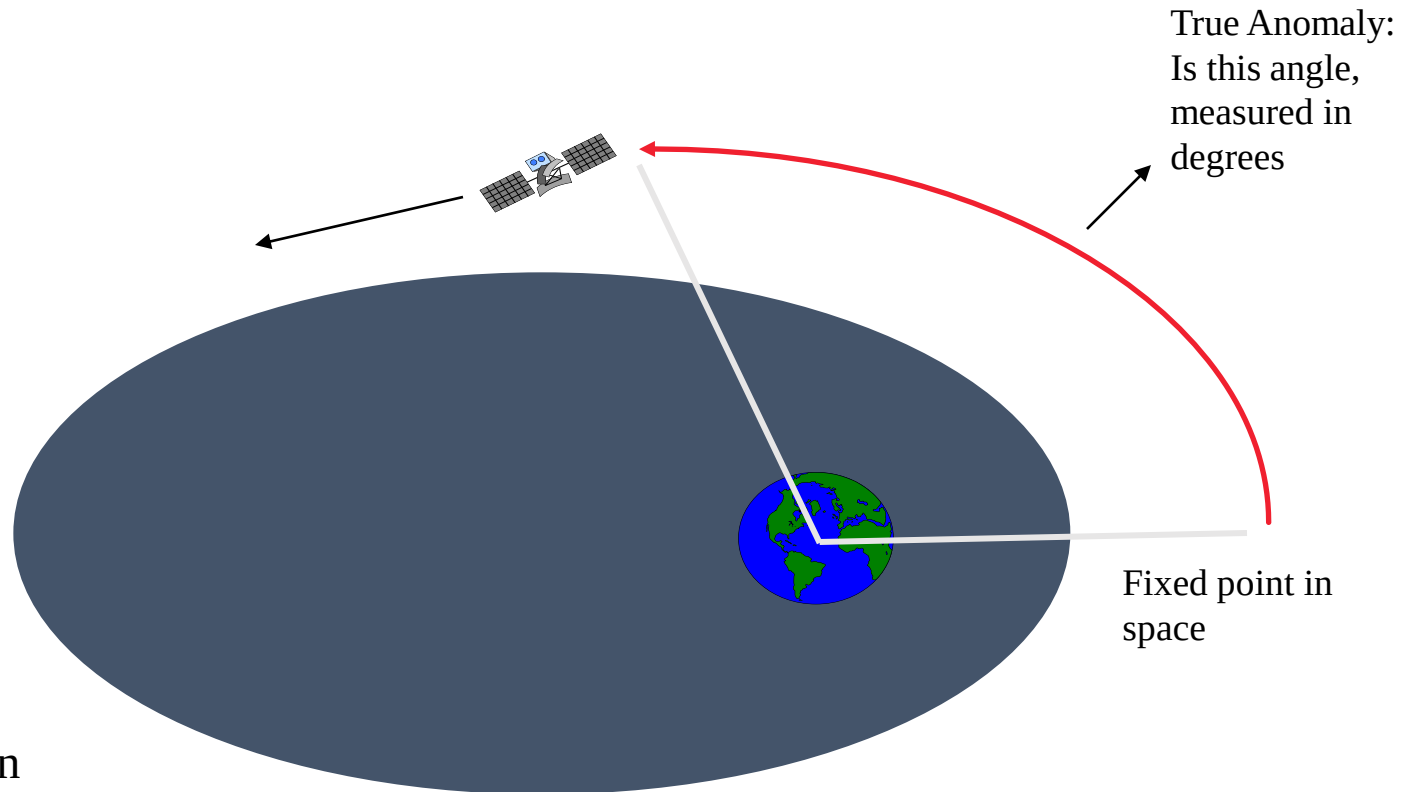
- Argument of Perigee is a measurement from a fixed point in space to where perigee occurs in the orbit
- It is measured in degrees



ORBITAL ELEMENTS

True Anomaly

- True Anomaly is a measurement from a fixed point in space to the actual satellite location in the orbit
- It is measured in degrees



You will be able to much easily see what True Anomaly is when using STK!!

GROUND TRACKS!!

GROUND TRACKS

Definition

- One way to define a satellite's orbit is to determine its track across the ground
- It is as if you had a big pencil from the satellite to the ground. The track it traces is called the ground track

GROUND TRACKS

Definition

- Sub point
 - Point on Earth's surface defined by an imaginary line connecting the satellite and the Earth's center
- Ground Track
 - Trace of sub points over time

GROUND TRACKS

Factors

- Size/Period
- Eccentricity
- Inclination
- Argument of Perigee
- Injection Point

Ground Tracks

Period

- For a non-rotating Earth, the ground track of a satellite is a great circle**
- Since the Earth spins on its axis and the satellite orbits the Earth, the period of both affects the ground track**

Ground Tracks

Westward Regression

- **Earth rotates east under a satellite => satellite appears to walk west**
- **Earth rotates 360 degrees in 24 hours (15 degrees per hour)**

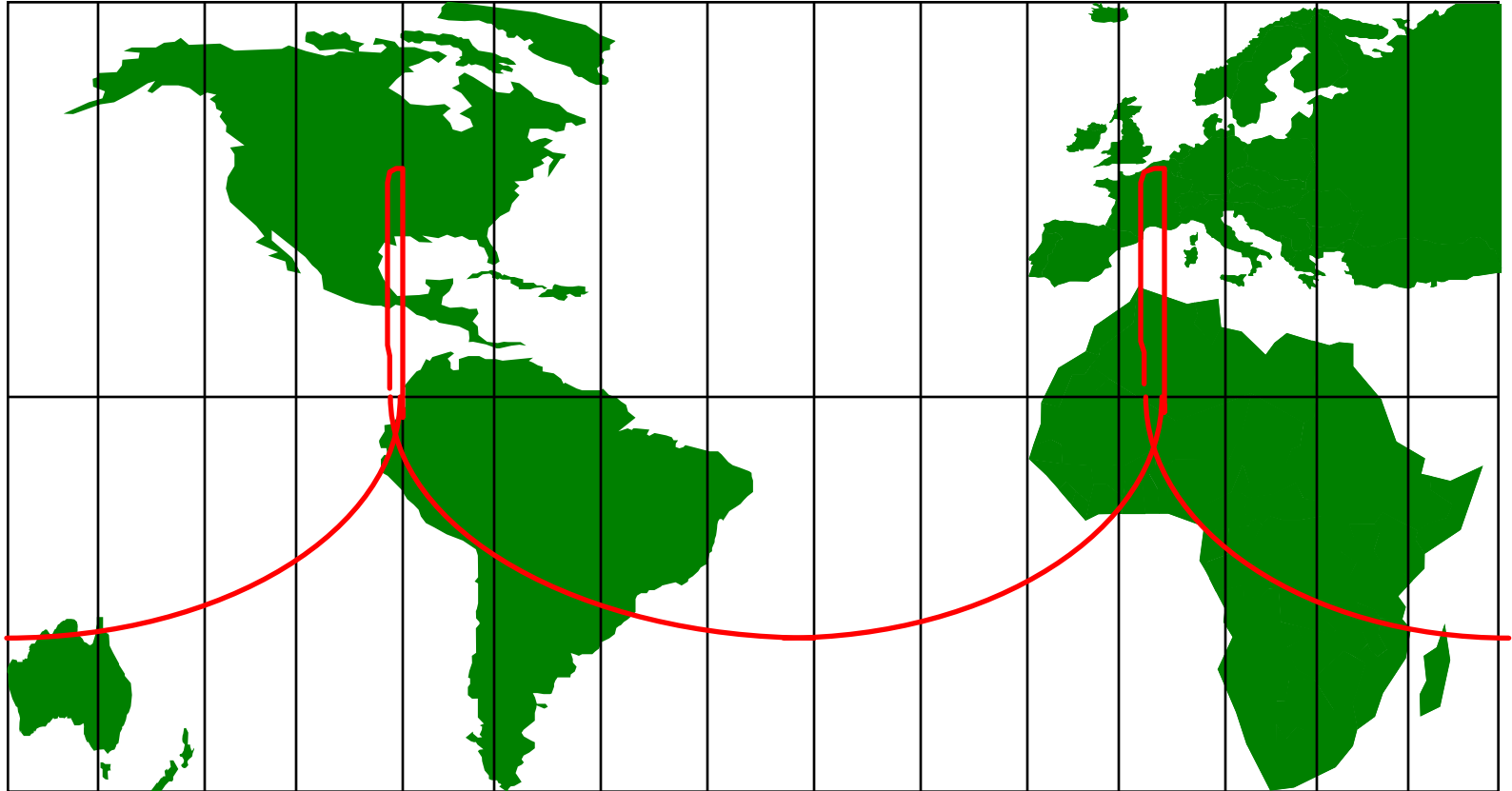
Ground Tracks

Eccentricity

- Highly eccentric orbit means satellite moves faster at perigee and slower at apogee => ground track will be asymmetrical
- Satellite will 'hang' over earth at apogee and move faster than the earth at perigee

Ground Tracks

Eccentricity



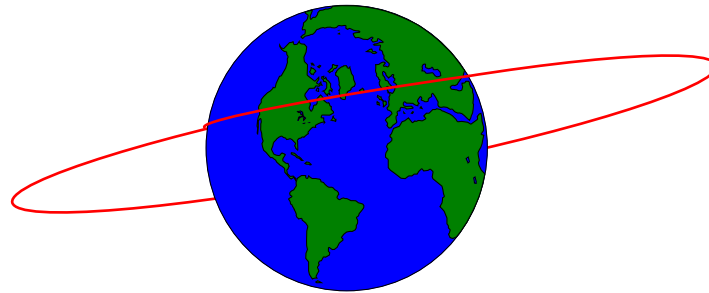
Ground Track for Molnya orbit
eccentricity = .7252



Ground Tracks

Inclination

- Inclination of the orbit determines the maximum latitude the ground track will reach



Ground tracks

Inclination



Inclination = 45 degrees

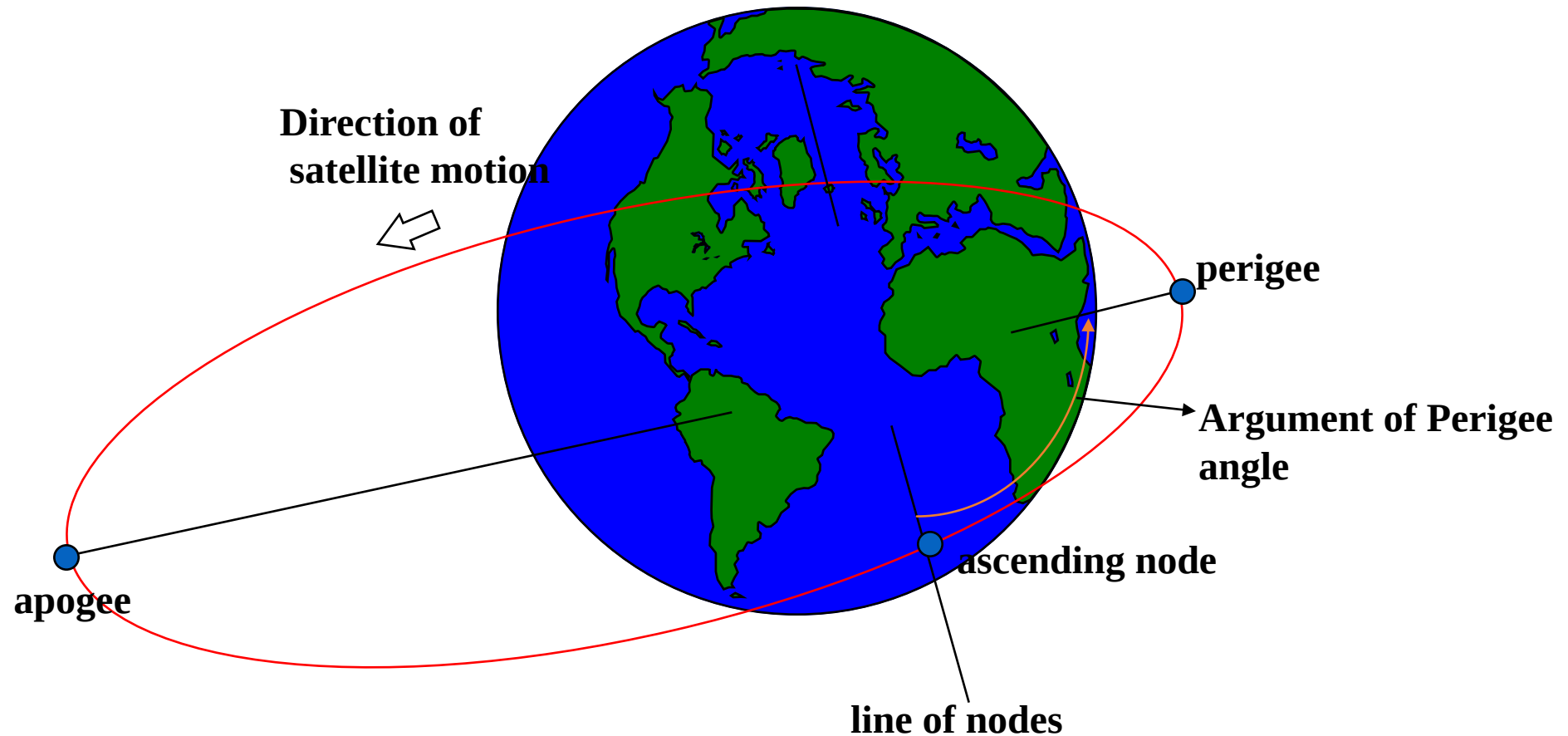
Eccentricity ~ 0



Ground Tracks

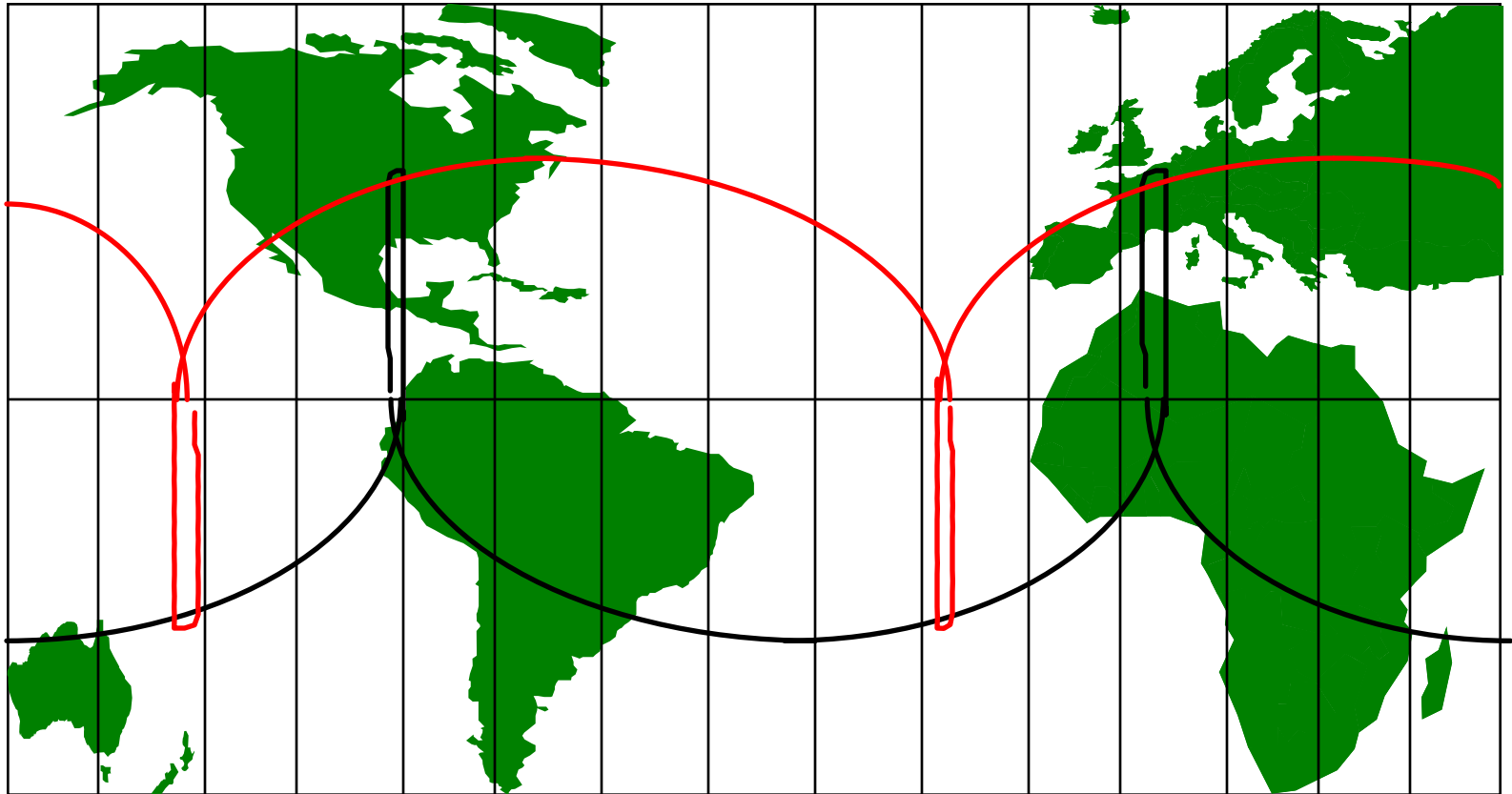
Argument of Perigee

- Establishes the longitude of both perigee and apogee



Ground tracks

Argument of Perigee



Argument of Perigee ~ 90 degrees (red)
argument of perigee ~ 270 degrees (white)



Ground tracks

Injection Point

- Assuming no maneuvers after launch, launch sites will determine inclination - more on this in launch considerations
- Injection point will determine where the ground track will start

PERTURBATIONS

- Space is a vacuum
- Once a satellite is in orbit, in the vacuum of space, is there anything that will affect it??
- Yes – these things are called Perturbations.....

PERTURBATIONS

- Definition
 - A disturbance in the regular motion of a celestial body
- Types
 - Gravitational
 - Atmospheric Drag
 - Third Body Effects
 - Solar Wind/Radiation Effects
 - Electro-magnetic

PERTURBATIONS

Gravitational

- Earth's asymmetrical mass causes a non-central gravitational pull

PERTURBATIONS

Gravitational

- Ellipticity of the Earth causes gravity wells and hills
- Stable points: 75E and 105W
 - Himalayas and Rocky Mountains
- Unstable points: 165E and 5W
 - Marshall Islands and Portugal
- Drives the requirement for station keeping

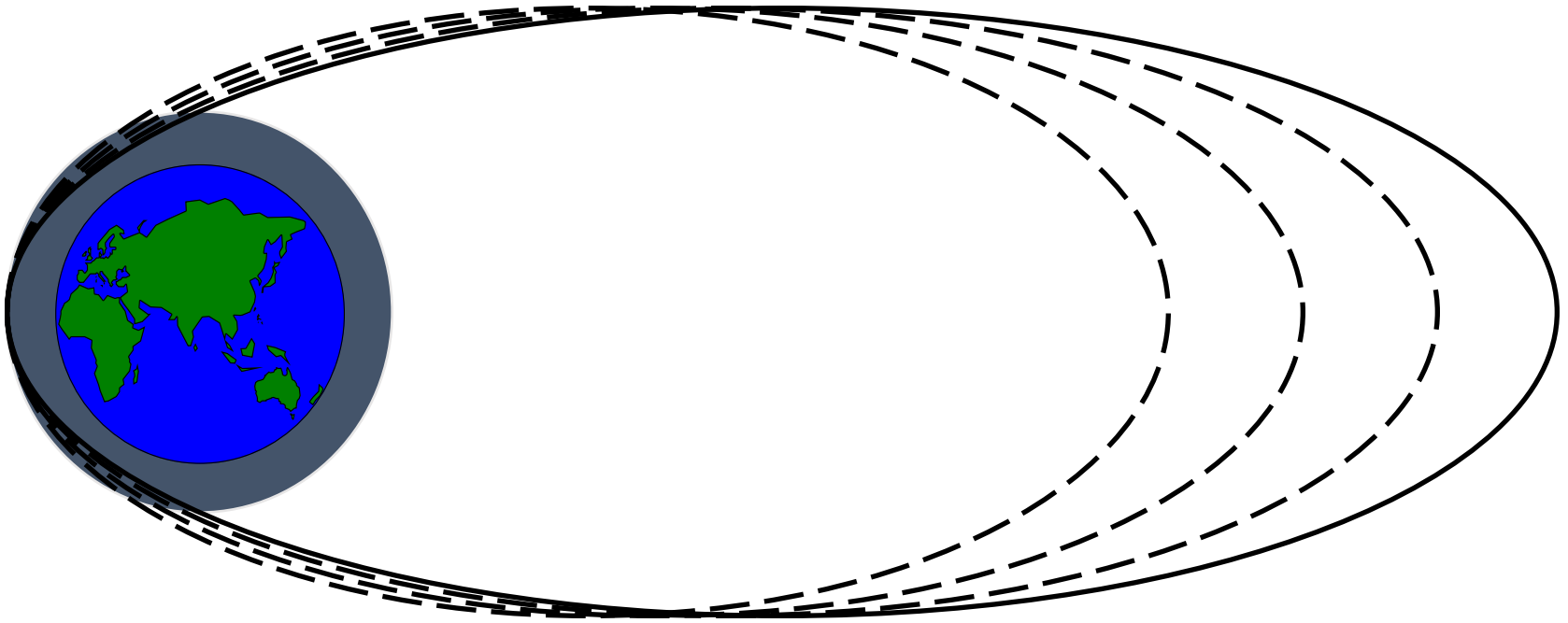
PERTURBATIONS

Atmospheric Drag

- Friction caused by impact of satellite with particles in the Earth's atmosphere
- Reduces satellite's energy
- Changes the size (semi-major axis) and shape (eccentricity)

PERTURBATIONS

Atmospheric Drag

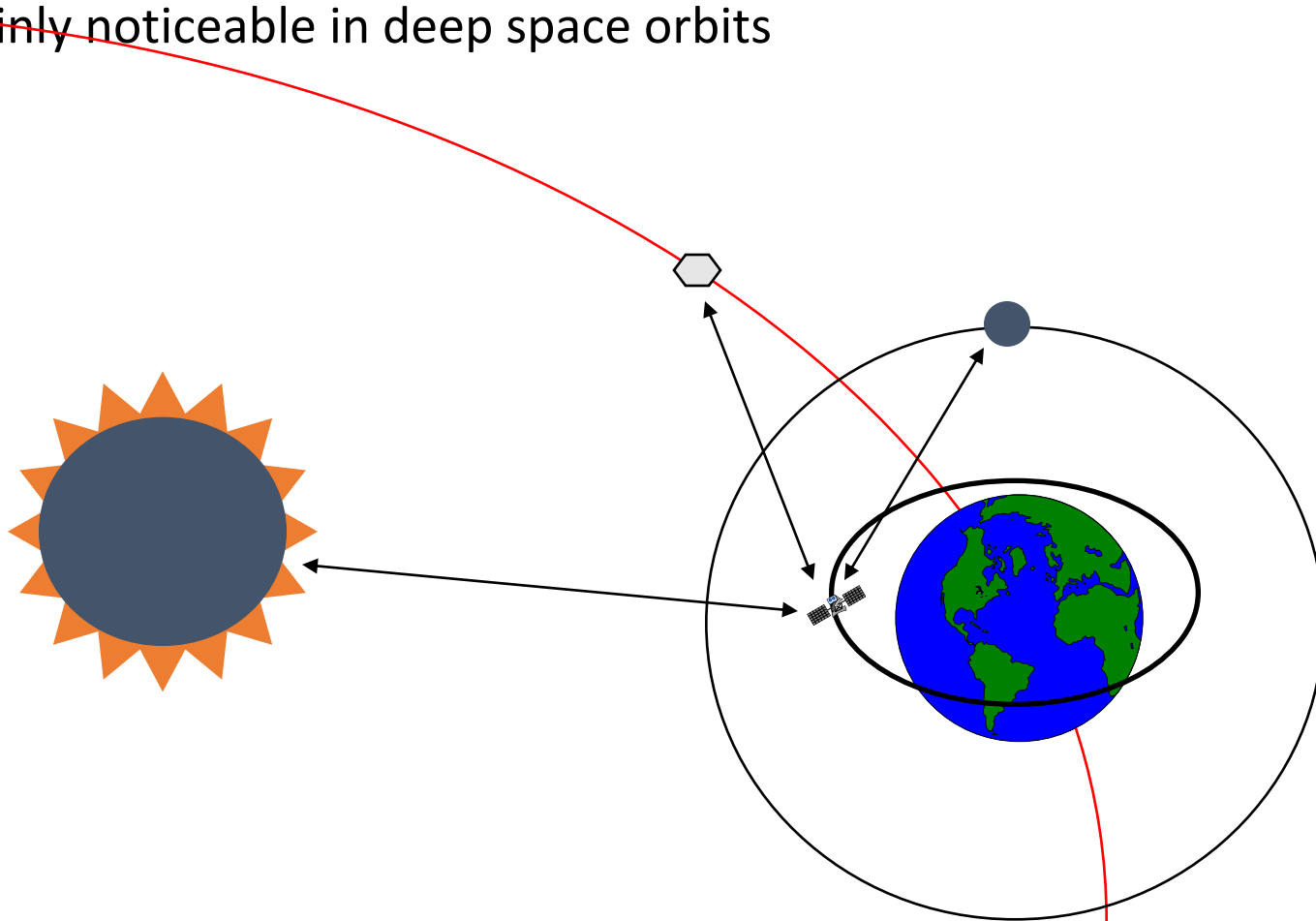


Perigee remains same, Apogee decreases

PERTURBATIONS

Third Body Effects

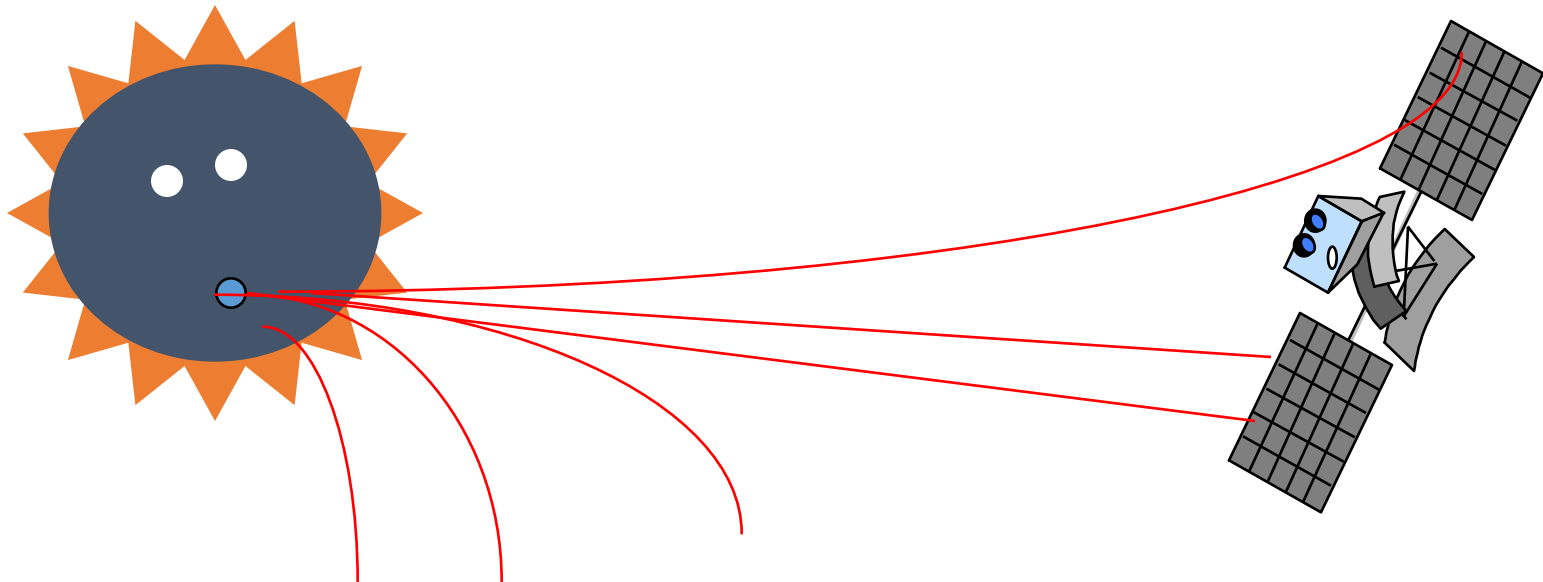
- Gravitational pull of other massive bodies, i.e. Sun, moon
- Mainly noticeable in deep space orbits



PERTURBATIONS

Solar Wind/Radiation Pressure

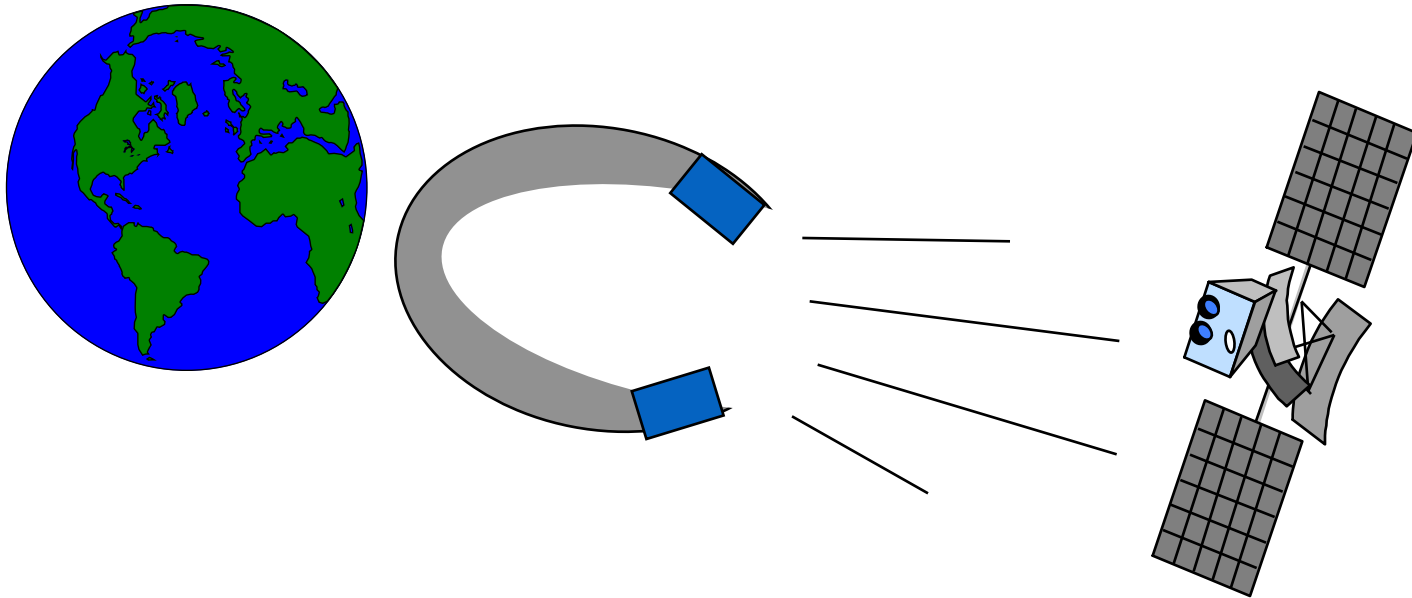
- Solar wind causes radiation pressure on the satellite
- Effects similar to atmospheric drag
- Effects are more pronounced on satellites with large surface areas



PERTURBATIONS

Electro-Magnetic

- Interaction between the Earth's magnetic field and the satellite's electro-magnetic field results in magnetic drag



LAUNCH CONSIDERATIONS

- Launch Windows
- Azimuth Vs. Inclination

LAUNCH CONSIDERATIONS

Launch Windows

- The period of time during which a satellite can be launched directly into a specific orbital plane from a specific launch site
- Window duration driven by safety, fuel requirements, desired injection points, etc.
- Window is centered around optimal launch time

LAUNCH CONSIDERATIONS

Launch Windows

- Opportunities to launch DIRECTLY into orbital plane
 - 2 per day if latitude of launch site is less than orbit's inclination
 - 1 per day if latitude is equal to inclination
 - None if latitude is greater than inclination

LAUNCH CONSIDERATIONS

Azimuth Vs. Inclination

- Launching due east, or at an azimuth of 90 degrees will result in an orbital inclination equal to launch site latitude
- Any other azimuth results in a GREATER inclination
- Azimuth selected for initial velocity boost and to avoid populated areas
- Proper azimuth minimizes future plane change requirements

ORBITAL MANEUVERS

- Reasons
- Types
- Methods

ORBITAL MANEUVERS

Reasons

- Maneuver to higher orbit
 - Increase satellite Field-of-view (FOV)
 - Counteract atmospheric effects
- Maneuver to lower orbit
 - Increase imaging resolution
 - Satellite rendezvous
 - De-orbit

ORBITAL MANEUVERS

Types

- In-plane
 - Change in size/period
 - Change in argument of perigee
 - Change in true anomaly
- Out-of-Plane
 - Change in inclination
 - Change in RAAN

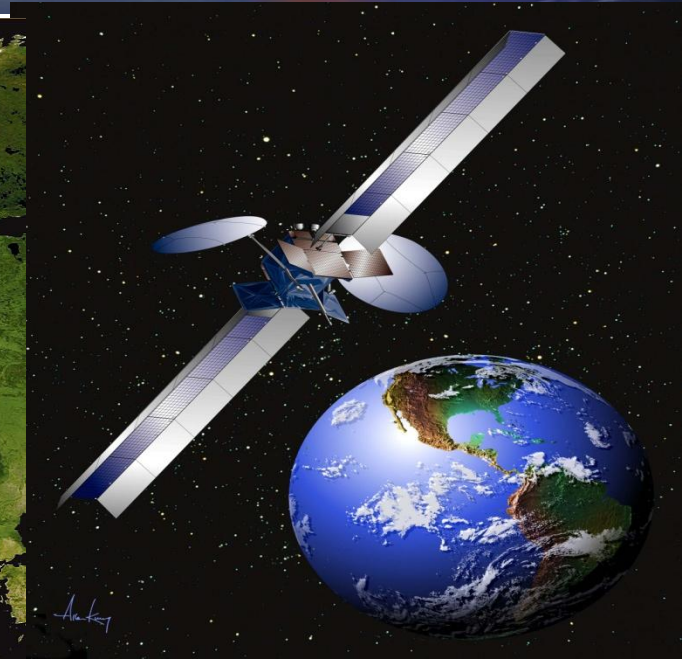
DE-ORBIT/DECAY

- De-Orbit is the controlled re-entry of a satellite to a specific location
 - Used for the recovery of payload
 - Manned missions
- Decay is uncontrolled re-entry
 - Potential impact anywhere along ground track
 - Re-entry Assessment (by CMAS)



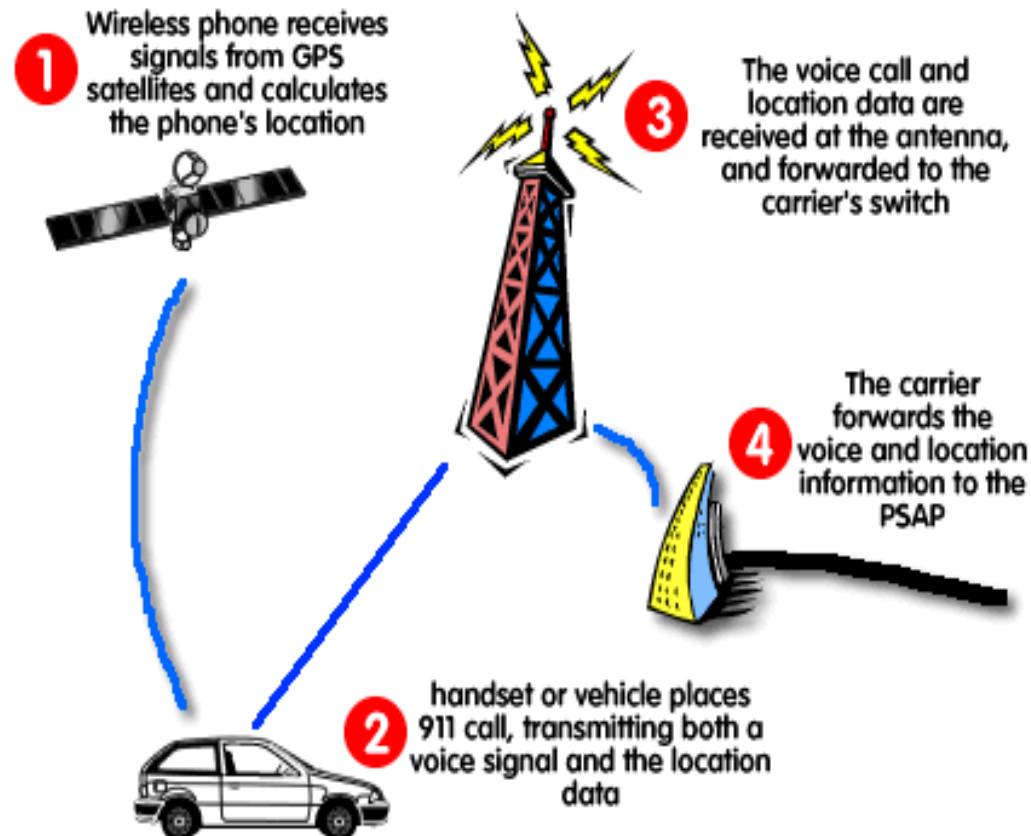
TYPES OF ORBITS - Uses of Satellites

- Daily Uses of Satellites
- Big Picture
- Affects of Altitude



TYPES OF ORBITS - Uses of Satellites

Global Positioning System!!



TYPES OF ORBITS - Uses of Satellites

A Remote Sensing Satellite's view of Earthquake Damage in Haiti



PLACING SATELLITES IN ORBIT OVERVIEW

- How Satellites are Launched
- Location Advantages of the Two Primary US Launch Site

PLACING SATELLITES IN ORBIT

- You need LIFT !!

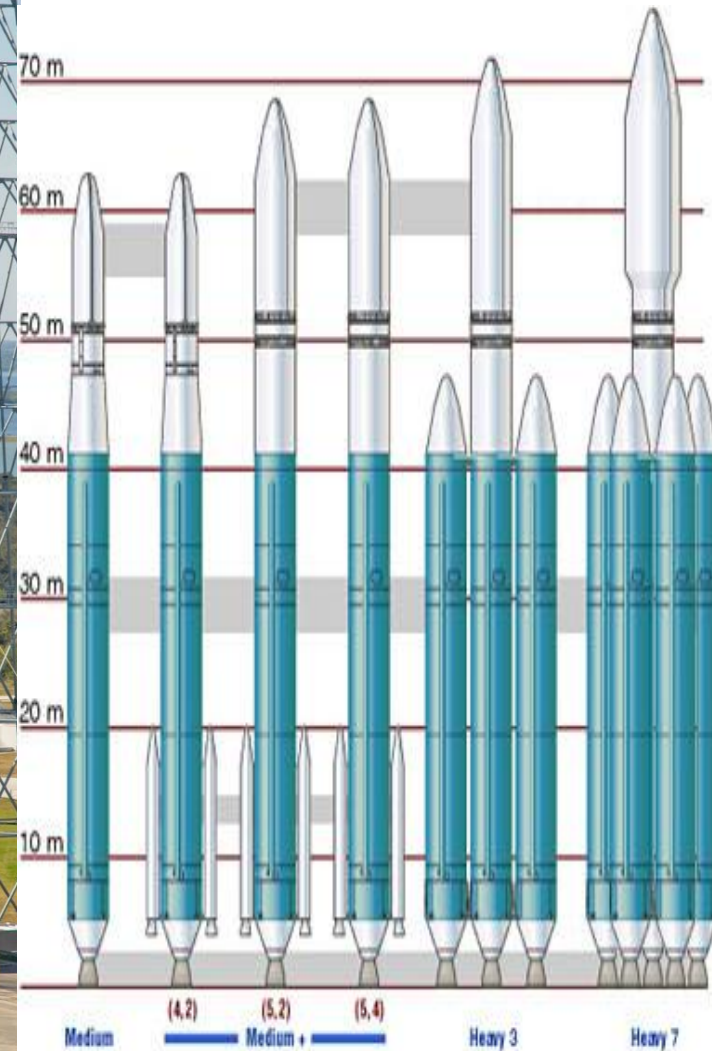
$$W = m (g)$$

Weight = mass (acceleration of gravity)

PLACING SATELLITES IN ORBIT

- Boosters

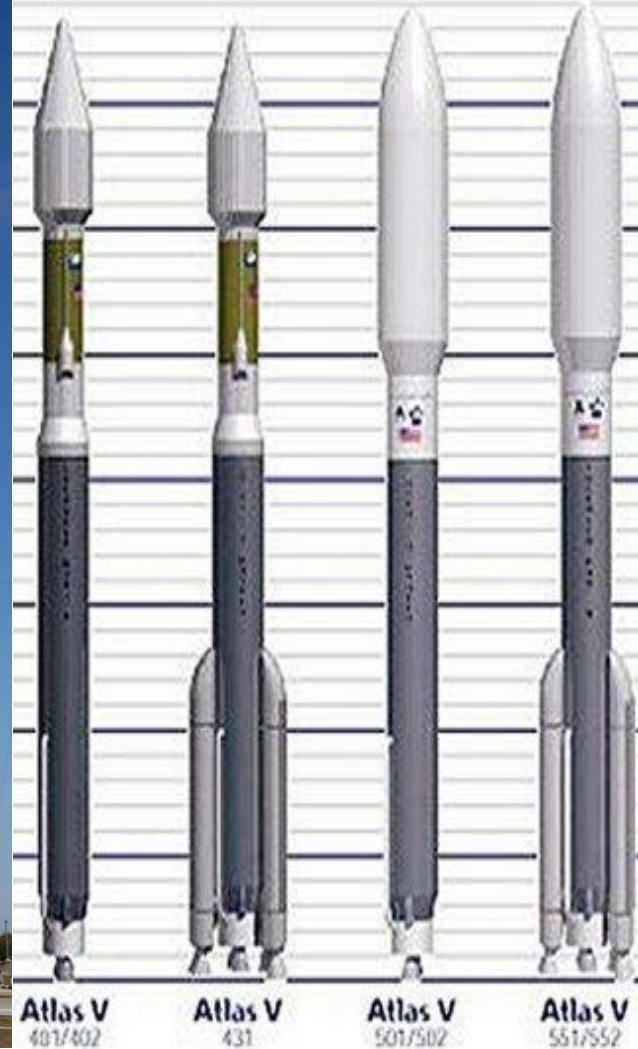
DELTA IV



PLACING SATELLITES IN ORBIT

- Boosters

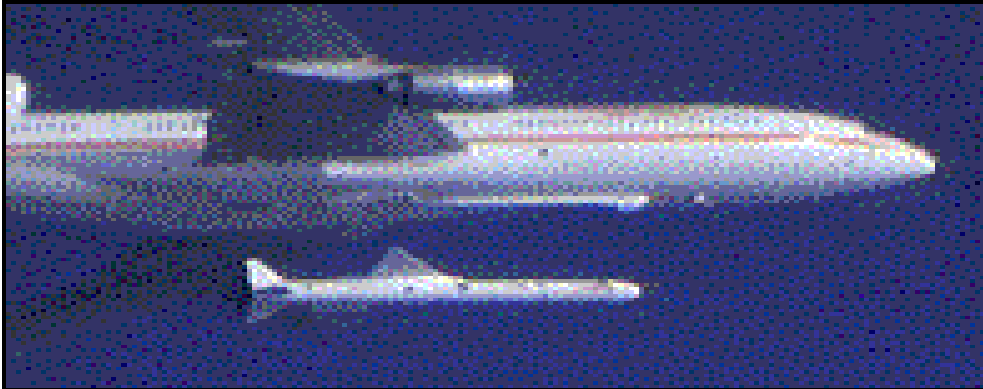
ATLAS V



PLACING SATELLITES IN ORBIT

- Boosters

PEGASUS



NASA Dryden Flight Research Center Photo Collection

<http://www.dfrc.nasa.gov/gallery/photo/index.html>

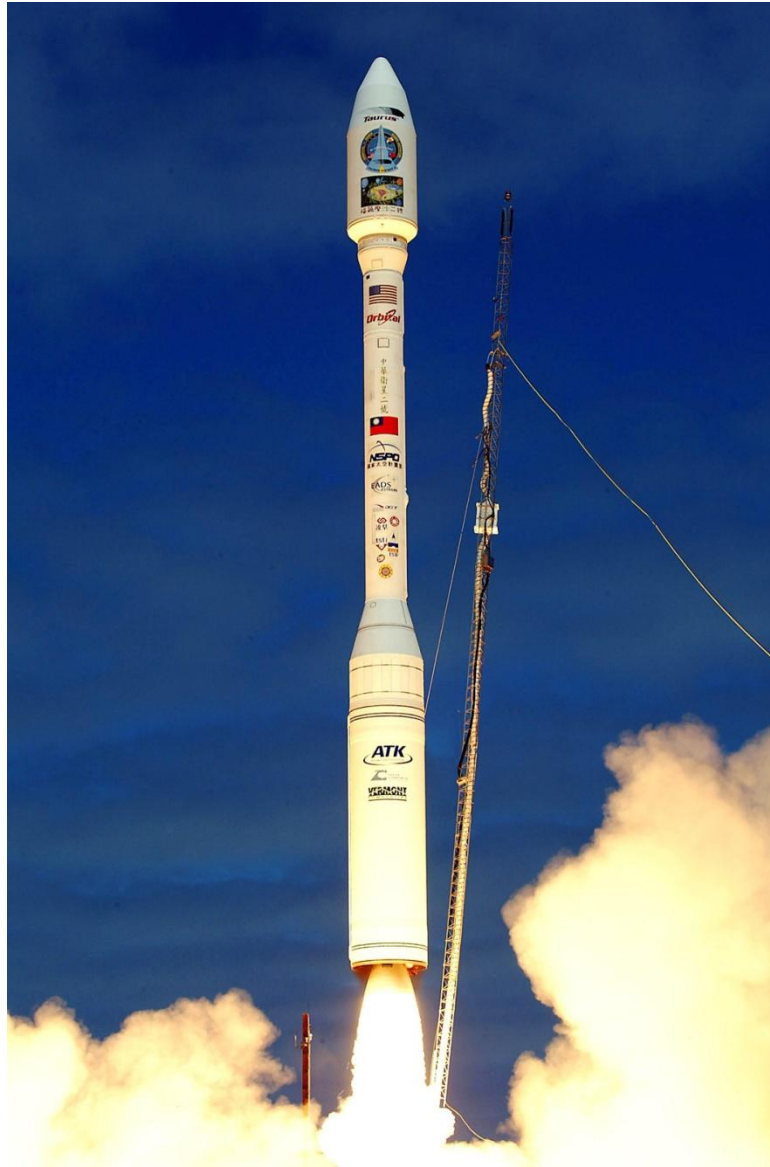
NASA Photo: EC01-0182-35 Date: June 2, 2001 Photo by: Jim Ross

Ignition of the Pegasus rocket moments after release from the B-52 signaled acceleration of the X-43A/Pegasus combination over the Pacific Ocean.

PLACING SATELLITES IN ORBIT

- Boosters

TAURUS



PLACING SATELLITES IN ORBIT

- Boosters

The
SHUTTLE
BOOSTER



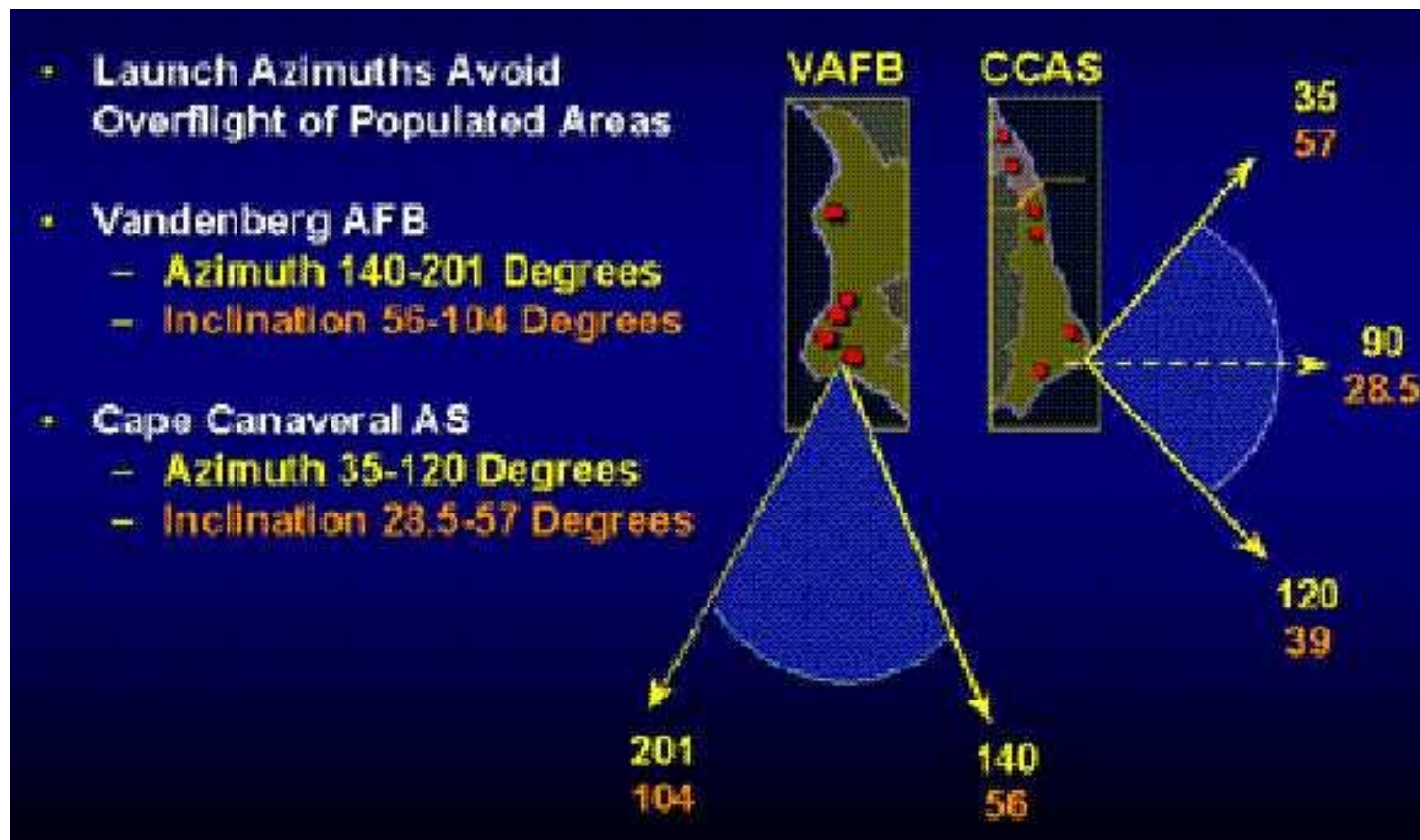
PLACING SATELLITES IN ORBIT

- **Launch Locations**

- Cape Canaveral (Patrick AFB) Eastern Range
- Vandenberg AFB (Western Range)

PLACING SATELLITES IN ORBIT

- Launch Constraints

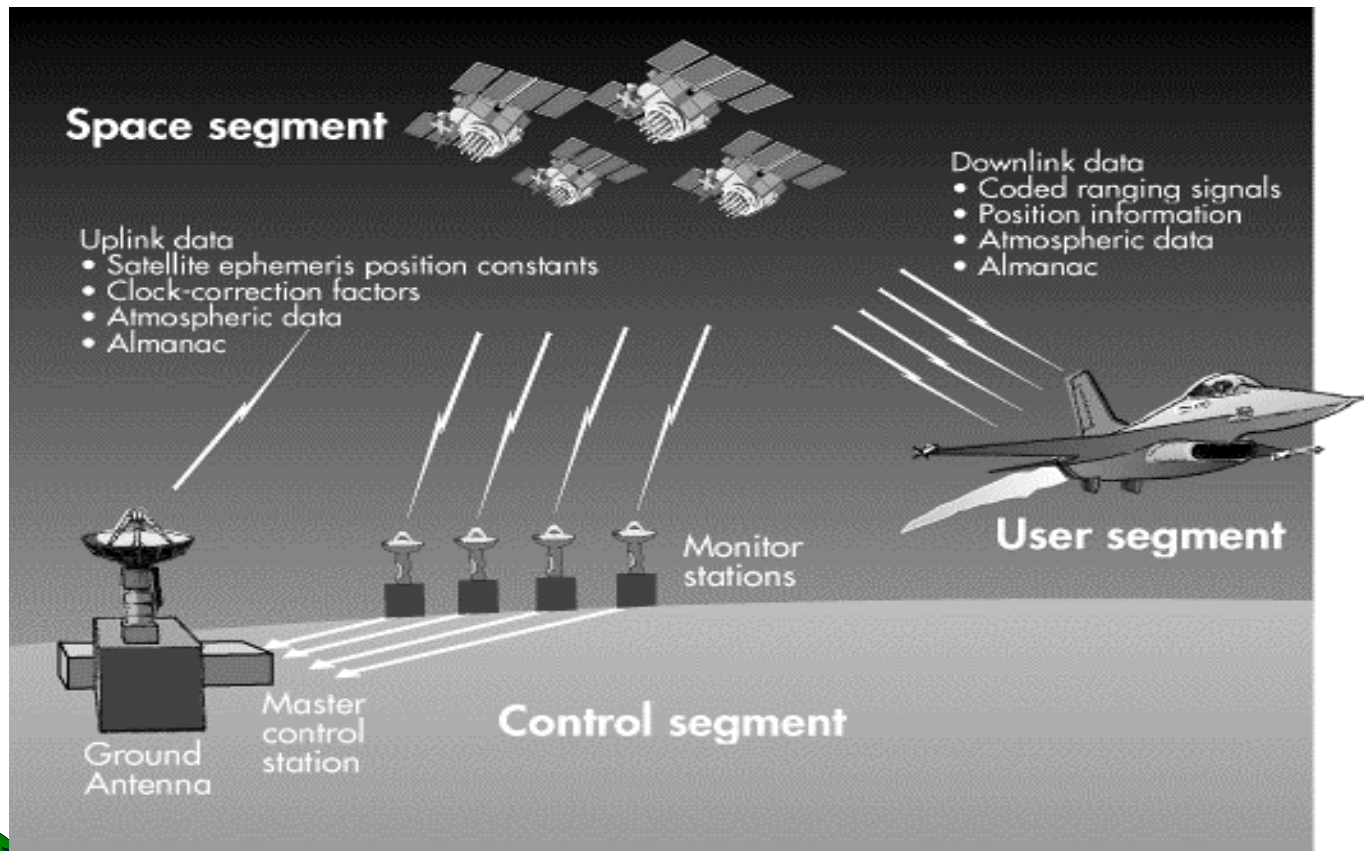


SATELLITE OPERATIONS *ELEMENTS*

- Ground Segment
- Space Segment
- Data Link Segment

SATELLITE OPERATIONS *FUNCTIONS*

- GPS Example



SATELLITE OPERATION *ACCESS*

- Field of View (FOV)
- Location of Ground station/Observer
- Satellite Orbital Position

ORBITAL MECHANICS

- Classroom Presentations
using Powerpoint
- Demonstrate with STK
- Let's Demo !!



The world of Space Operations awaits you!!